

SIRT2 protects against Japanese encephalitis virus infection in mice

Reviewed Preprint

v1 • June 27, 2025

Not revised

Perumal Arumugam Desingu , Lavanya Dindi, Krishnega Murugasamy, Ankit Kumar Tamta, Venkatesubbu Ramasubbu, Sukanya Raghu, Amarjeet Shrama, Raju S Rajmani, Nagalingam R Sundaresan 

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India • Department of Virus Epidemiology, Vector Dynamics & Public Health, Institute of Advanced Virology, Trivandrum, India • Centre for Infectious Disease Research, Indian Institute of Science, Bengaluru, India

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **important** study explores the role of SIRT2 in regulating Japanese encephalitis virus replication and disease progression in rodent models. The findings presented are novel as sirtuins are known for their roles in aging, metabolism, and cell survival, but have not been studied in the context of viral infections until recently. The evidence supporting the claims is **solid**, although additional experiments to further characterize the clinical outcomes and directly test the link between acetylated NF- κ B and SIRT2 expression would have strengthened the study. The work will be of interest to biologists studying viruses, sirtuins, and inflammation.

<https://doi.org/10.7554/eLife.106778.1.sa3>

Abstract

Japanese encephalitis virus (JEV) is a mosquito-borne zoonotic RNA virus that causes Japanese encephalitis (JE) and poses a major threat to public health in Southeast Asia and the Western Pacific. Current strategies rely on prophylactic methods to prevent disease, as no effective antiviral therapy exists. Here, we report that SIRT2, an NAD⁺-dependent deacetylase enzyme, mediates antiviral activity against JEV infection in mice. Interestingly, our study reveals that SIRT2 is downregulated in JEV infection, SIRT2 genetic deficiency/small molecule inhibition increases viral yield in neuronal cells and mice brains thereby reducing the survival rate in the infected mice, whereas SIRT2 gene therapy to the JEV-infected mice by Adeno-associated virus vector reduced the JEV load in mice brains and improved the survival rate. SIRT2 deficiency activates inflammatory cytokines and chemokines response in the JEV-infected mice brains through activating NF- κ B transcription factor. Mechanistically, SIRT2 deacetylates NF- κ B to reduce the transcriptional factor activity of NF- κ B that down-regulates the Beclin-1-mediated autophagy, which is needed for the JEV replication. Overall, the present findings establish SIRT2 as a potential regulator of JEV infection.

Introduction

JE is a vector-borne zoonotic viral disease caused by the Japanese encephalitis virus (JEV)^{1,2,3}. JEV exhibits close resemblance to other members of the Flaviviridae family, including dengue virus, Zika virus, yellow fever virus, and West Nile virus^{1,4}. Estimates suggest that JE is endemic in about twenty-four countries in the Southeast Asia, South Asia, and Western Pacific geographic regions. Almost 68 thousand clinical cases occur every year, posing a risk to three billion people^{1,3,5}. In the endemic areas, children aged 2 to 15 years are more susceptible to JEV, and exhibit adverse outcomes, with a 30% fatality rate and permanent neurologic or psychiatric sequelae of 30-50% in encephalitis patients^{1,3,5}. Currently, vaccine administration in the endemic area is the only intervention against JEV¹. The absence of clinically available therapeutics raises serious health threats and warrants the identification of master regulators of host proteins in the virus pathogenesis to determine an antiviral against JEV. Because of the high rate of mutations in RNA viruses⁶⁻⁹, it would be appropriate to target host proteins and the evolutionarily conserved host mechanisms for therapeutic intervention.

In this line, we recently identified the inhibition of NAD⁺-dependent host enzyme, poly (ADP-ribose) polymerases (PARP1) inhibition, as a potential antiviral strategy¹. Possibly other NAD⁺-dependent host enzymes may also play a role in the JEV infection and the identification of such enzymes may be useful in designing a host protein-directed antiviral against JEV infection. In this context, sirtuins (SIRT) are the well-explored NAD⁺-dependent enzymes and sirtuins' role in RNA virus-infected animals/humans is largely unknown. Sirtuins are class III histone deacetylases that depend on NAD⁺ as a cofactor for their enzymatic activity. Mammals have seven sirtuins (SIRT1-SIRT7), ubiquitously expressed with distinct sub-cellular localisation^{10,11}. SIRT governs several cellular processes through multiple targets and have critical implications in health and diseases¹²⁻¹⁴. Of them, the major cytosolic protein SIRT2 is known to regulate cell division, inflammation, autophagy, mitophagy, oxidative stress, and senescence^{11,15,16}. In this context, the role, underlining molecular mechanism and signalling pathways of cytosolic sirtuin, SIRT2, in RNA viruses' multiplication and viral pathogenesis in animal models remains largely unknown. Understanding the role of SIRT2 in distinct viral pathology, especially encephalitis-causing RNA viruses in mice models, and exploring molecular mechanisms in the evolutionarily conserved signalling axis, holds direct therapeutic relevance for preclinical studies. Further, this knowledge can be extended to develop therapeutics against several other viruses that utilise the same host proteins and similar mechanisms for their pathogenesis.

Several RNA viruses including the hepatitis C virus (HCV)^{17,18}, porcine reproductive and respiratory syndrome virus (PRRSV)^{19,20}, Severe acute respiratory syndrome coronavirus (SARS-CoV)²¹, influenza A virus (IAV)^{22,23}, JEV¹, Dengue virus (DENV)^{24,25}, Zika virus (ZIKV)^{26,27}, Chikungunya virus (CHIKV)²⁸, Human rhinovirus (HRV)²⁹, Enterovirus 71 (EV-A71)³⁰, poliovirus^{31,32}, Coxsackievirus^{33,34}, encephalomyocarditis virus (EMCV)³⁵, Foot-and-mouth disease virus (FMDV)³⁶, Newcastle disease virus (NDV)³⁷ etc., exploit evolutionarily conserved cellular autophagy for viral replication by different molecular mechanisms. In this context, the functional role of SIRT2 in RNA viruses-induced autophagy has not yet been explored. Of note, different RNA viruses target the host protein nuclear factor- κ B (NF- κ B), to modulate the inflammatory cytokine and chemokine transcription (reviewed in³⁸⁻⁴¹). The role of phosphorylation of I κ B in the NF- κ B-I κ B axis^{42,43} has been well-studied in viral infections. In contrast, the role of post-translational modifications, particularly acetylation of NF- κ B, on its transcriptional factor activity-mediated regulation of RNA virus-induced autophagy remains unexplored.

In our study, we uncover a key role for SIRT2 in regulating JEV replication and improving the survival in JEV-infected mice. Our *in vitro* and *in vivo* model systems demonstrate that this effect is SIRT2-mediated deacetylation of NF- κ B thereby regulating virus-induced autophagy and inflammatory cytokine. Further, we report that SIRT2 mediated NF- κ B regulation negatively affects the virus-induced autophagy, which is needed for virus replication. Thereby SIRT2 exhibits an antiviral role in neuronal cells and mice infected with JEV.

Results

The JEV infection down-regulates SIRT2 in the *in vitro* and *in vivo* models

Our recent study determined the role of NAD⁺-dependent enzyme PARP1 in JEV replication¹. However, the role of other NAD⁺-dependent enzymes remains unexplored. We were interested in exploring the role of NAD⁺-dependent deacetylase enzymes, Sirtuins⁴⁴, in JEV replication. Because JEV replicates in the cytoplasm^{1,3,5}, we wanted to understand the importance of SIRT2, a cytoplasmic sirtuin^{10,11}, in the regulation of JEV replication. To begin with, we aimed to identify the molecular mechanisms occurring at the early viral replication time point that can be targeted as an antiviral against JEV infection. To determine the same, we infected the neuro2a cells with one multiplicity of infection (MOI) of JEV and observed the viral protein levels and infective virus yield over time. We found that the maximum rate of JEV multiplication occurs between 24-36 hours post-infection (hpi) (Figure 1A–1B) and therefore considered 30 hpi as an early accelerating phase of viral replication (Figure 1B). At this time point, SIRT2 was significantly down-regulated (Figure 1C).

Further, to investigate the role of SIRT2 in an *in vivo* system, we studied the brains of JEV-infected mice. Since JEV infection is spread through mosquito bites and causes encephalitis in young children⁴⁵, we infected 10 days old mice pups with 10⁵ plaque-forming units (PFU) of JEV by foot-pad route to mimic natural infection. The infected mice were monitored for the display of clinical signs, paralysis, and survival every 24 hours post-infection. The standard clinical signs observed include fluffy coat, ruffled fur, reluctance to move, hunchback posture, circling and curling behaviour, ataxia, and tremors on the 6th day post-infection (dpi). By the 11th dpi, we observed that about 90 per cent of animals displayed clinical signs (Figure 1D). The mice infected with JEV started showing paralysis at the 8th dpi, which peaked by the 13th day (Figure 1E). We observed mortality in JEV-infected mice 10th dpi onwards (Figure 1F). The serum viral load was detectable 3rd dpi onwards and peaked on the 6th dpi, coinciding with the onset of clinical signs (Figure 1D, 1G). Interestingly, serum viral load started declining when the virus started multiplying in the mice's brain (Figure 1G, 1H). The maximum rate of viral replication in the brain, primary target site of JEV, occurred between 7 and 9th dpi (Figure 1H). Thus, 8th dpi was noted to be an early accelerating phase of viral replication in the brain of mice (Figure 1H). Next, we measured the SIRT2 levels on 8th dpi. In line with our *in vitro* findings, SIRT2 levels were significantly downregulated at 8th dpi (Figure 1I). Thus, significant SIRT2 downregulation coincides with the early accelerating phase of viral replication in both *in vitro* and *in vivo* study systems.

SIRT2 deficiency enhances JEV replication in both *in vitro* and *in vivo* models

Since SIRT2 is downregulated in JEV infection, we infected SIRT2-depleted neuro2a cells with 1 MOI JEV to study the role of SIRT2 in viral multiplication. We found that viral titre was significantly high in SIRT2 knockdown conditions (Figure 2A, 2B). In contrast, SIRT2 overexpression remarkably reduced viral titre in neuro2a cells (Figure S1A, S1B), thus suggesting a role for SIRT2 in JEV multiplication. To validate our findings in the *in vivo* model system, we

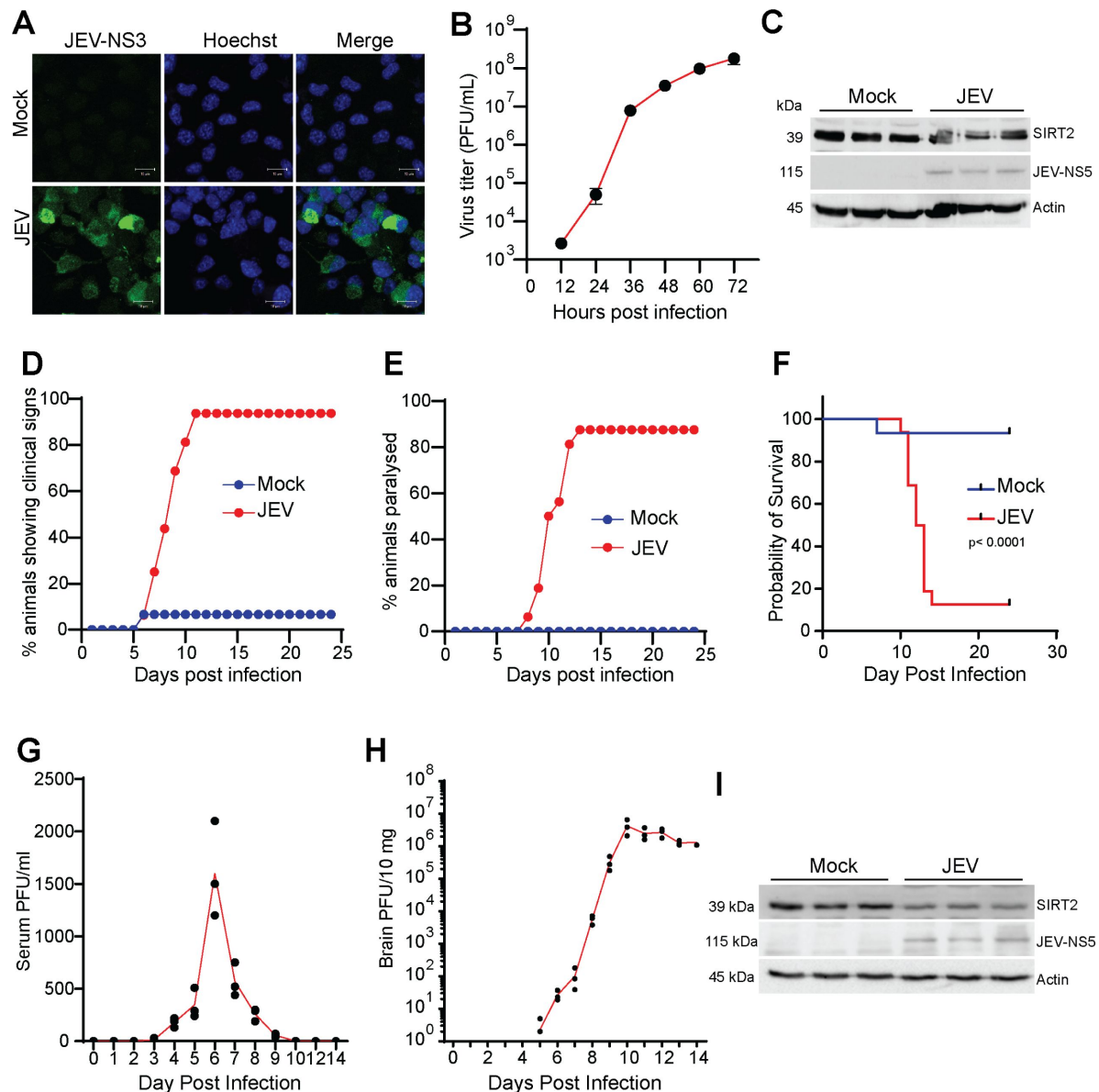


Figure 1.

SIRT2 downregulated in neuro2a cells and mice brains upon JEV infection.

(A) The representative confocal images of JE virus (one MOI) or mock-infected neuro2a cells at 30 hpi; JEV-NS3 protein stained in green, and the nuclei stained by Hoechst 33342 in blue. The scale bar = 10 mM. (B) The virus load upon 1 MOI of JE virus infection in neuro2a cells over time expressed in plaque-forming units (PFUs). $n=3$ independent experiments; each experiment performed by duplicates. (C) Representative immunoblot image displaying a reduction in SIRT2 levels in neuro2a cells upon 1 MOI of JE virus infection at 30 hpi. The JEV protein NS5 confirms JEV infection in the cells. $n=3$ independent experiments; each experiment performed by duplicates. (D-I) 10-day-old mice were infected through the JE virus of 10^5 PFU by the foot-pad route of inoculation; $n=15-16$ mice per group. Graphical representation depicting the percentage of mock or JE virus-infected mice expressing clinical signs (D) and paralysis (E) at various dpi. (F) The percentage of mice surviving upon JEV mice at various dpi up to 24th dpi, in Kaplan-Meier survival percentage curve ($n=15-16$ mice each group). The p -value was calculated using the Log-rank (Mantel-Cox) test. Graphically representing the JEV load (PFUs) in the serum (G) and brain (H) of wild-type mice at various dpi; $n=3$ mice per dpi. (I) Representative immunoblot indicating the reduction in SIRT2 levels upon JEV-infection on 8th dpi. The detection of NS5 confirms the virus infection. $n=6$ mice in each group.

infected SIRT2 knock-out (KO) mice with JEV through the footpad route (**Figure 2C**). We observed that the viral PFU levels in serum reached a maximum in SIRT2-KO mice earlier than their WT counterparts (**Figure 2D–2E**). Similarly, the viral load in the JEV-infected SIRT2-KO mice brain was much higher than in WT (**Figure 2F**). Likewise, clinical signs and paralysis appeared earlier in SIRT2-KO mice than WT mice. The SIRT2-KO group also showed an increased percentage of animals exhibiting clinical signs and paralysis (**Figure 2G–2H**). Further, all the SIRT2-KO mice infected with JEV died by the 14th dpi, with significantly reduced survival of JEV-infected SIRT2-KO mice (**Figure 2I**). To ensure that the observed pathological signs are due to JEV infection in the brain rather than serum viral load, we employed the intracerebral route of JEV infection (**Figure 2J**). In corroboration with our findings from the foot-pad route of viral inoculation, the intracerebral route of viral infection also rendered SIRT2-KO mice more susceptible to JEV pathogenesis as indicated by increased viral titre, earlier onset of clinical sign, paralysis, and reduced survival rate (**Figure 2K–2N**). Thus, SIRT2 deficiency in the *in vitro* and *in vivo* models enhances JE virus replication thereby reducing the survival of JEV-infected mice, and SIRT2 overexpression helps curtail JEV titre *in vitro*.

Pharmacological inhibition of SIRT2 by AGK2 enhances JEV replication

Previous studies have shown that SIRT1–SIRT7 knockdown enhances a few RNA and DNA virus replication in cell culture models^{14,46,47}. In contrast, treatment with the SIRT2 inhibitor, AGK2 was shown to inhibit the replication of DNA viruses in mice models^{48,49}. Conversely, our study suggests that SIRT2 depletion enhances viral replication in mouse models. We next asked if SIRT2 inhibition, using the small molecule inhibitor AGK2, can enhance JEV replication. We treated neuro2a cells with AGK2 and observed significantly increased viral titers (**Figure 3A**). Further, we tested the effect of AGK2 in JEV infected mice. We found that JEV-infected mice, when treated with AGK2, showed early incidence and an increase in the percentage of clinical signs and paralysis (**Figure 3B, 3C**). AGK2 treatment also reduced the survival rate in JEV-infected mice (**Figure 3D**). We also observed an early appearance and increased serum viral load upon AGK2 treatment (**Figure 3E**). Similarly, the viral load in mice brain was elevated upon AGK 2 treatment (**Figure 3F, 3G**). Consistently, in mice infected via the intracerebral route, AGK2 treatment resulted in an early onset of clinical signs and paralysis (**Figure 3H, 3I**) and a reduction in survival rate compared to their mock infected counterparts (**Figure 3J**). Thus, SIRT2 inhibition using AGK2 exacerbated the clinical outcomes of JE in mice.

SIRT2 gene therapy protects against JEV infection in mice

Next, we were interested in exploring the therapeutic efficacy of SIRT2 gene therapy against JEV infection in mice. For this, we overexpressed SIRT2 in mock or JEV-infected mice brains using the Adeno-associated virus (AAV) system (**Figure 4A**). Interestingly, AAV-SIRT2 treatment after JEV infection had significantly less viral load as compared to AAV-Null treated, JEV infected mice brains (**Figure 4A, 4B**). In corroboration with reduced viral load in the JEV-infected WT mice brains, we also observed that SIRT2 gene therapy significantly reduced clinical signs and paralysis and enhanced survival rates in the JEV-infected wild-type mice compared to control AAV-vector treated JEV-infected mice (**Figure 4C–4E**). Overall, our results suggest that SIRT2 is a potential regulator of JEV infection in mice, and it could be used as a host protein-directed antiviral against JEV infection.

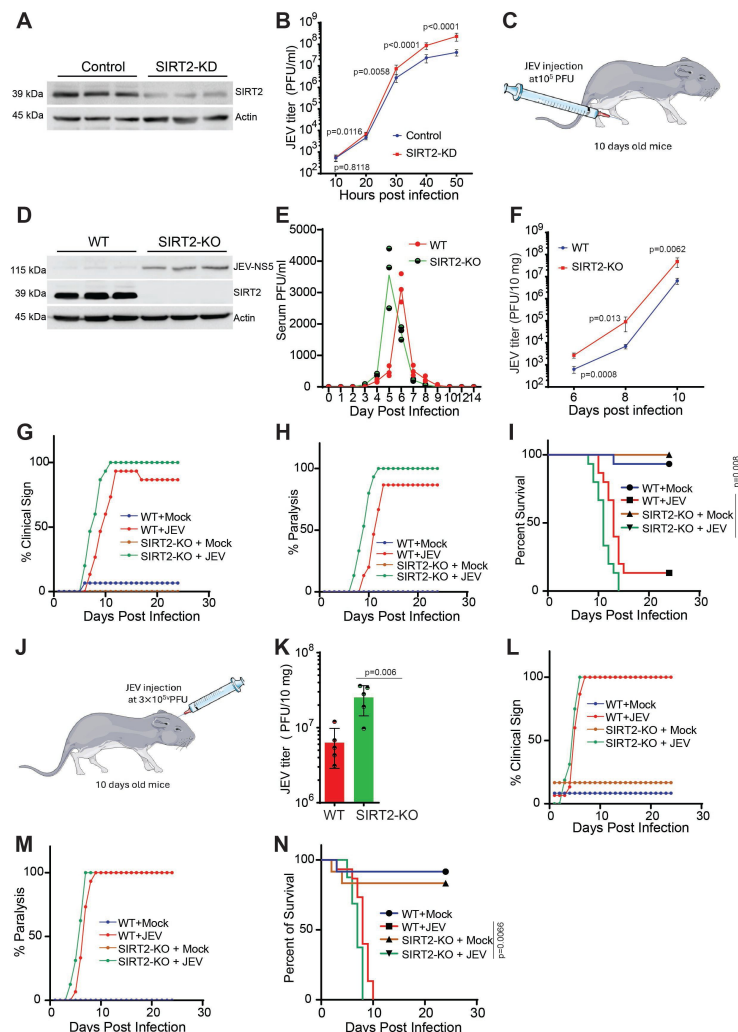


Figure 2.

SIRT2 genetic deficiency increases JEV multiplication in neuro2a cells and mice brains.

(A) Representative western blotting confirms SIRT2 knock-down in neuro2a cells by siRNA. (B) Graph displaying the viral PFU levels at the various hpi of control or SIRT2 knock-down neuro2a cells infected with JE-Virus. $n=3$ independent experiments; triplicates were used in each experiment. P values are calculated using Student's t-test. (C-I) 10-day-old mice were infected through the JE virus of 10^5 PFU by the foot-pad route of inoculation. (C) Schematic representation of JEV administration in mice by foot-pad route. (D) Representative immuno-blotting validating the absence of SIRT2 in the SIRT2^{-/-} (SIRT2-KO) mice brain; JE virus infection in the mice brain was confirmed by detecting JEV-NS5 protein. (E) The graph represents JE virus PFUs at various dpi in the serum of WT or SIRT2-KO mice infected with JE virus. $n=3$ mice per dpi. (F) The graph represents JE virus PFUs at various dpi in the brains of JE virus-infected WT or SIRT2^{-/-} mice. $n=5$ mice per dpi. P values are calculated using Student's t-test. (G) The graph represents the percentage of animals expressing clinical signs at various dpi in JE virus-infected WT or SIRT2^{-/-} mice ($n=15$ mice per group). (H) The graph depicts the percentage of animals displaying paralysis in JE virus-infected WT or SIRT2^{-/-} mice at various dpi ($n=15$ mice in each group). (I) The percentage of animals surviving in JE virus-infected WT or SIRT2^{-/-} mice at various dpi up to 24th dpi, in the Kaplan-Meier survival percentage curve. $n=15$ mice in each group; the p-value was calculated using the Log-rank (Mantel-Cox) test. (J-N) 10-day-old mice were inoculated with JE virus (3×10^3 PFUs) by an intra-cerebral route of inoculations. (J) Schematic representation of JEV administration in mice by an intra-cerebral route. (K) The graph represents JEV PFUs at 6th dpi in WT or SIRT2^{-/-} mice brains. $n=5$ mice per group; the p-value was calculated using Student's t-test. (L) The graph represents the percentage of animals expressing clinical signs at various dpi in JE virus-infected WT or SIRT2^{-/-} mice. $n=12-16$ mice in each group. (M) The graph depicts the percentage of animals displaying clinical paralysis at various dpi in JE virus-infected WT or SIRT2^{-/-} mice ($n=12-16$ mice in each group). (N) The percentage of animals surviving at various dpi up to 24th dpi in JE virus-infected WT or SIRT2^{-/-} mice in the Kaplan-Meier survival percentage curve. $n=12-16$ mice in each group; the p-value was calculated using the Log-rank (Mantel-Cox) test.

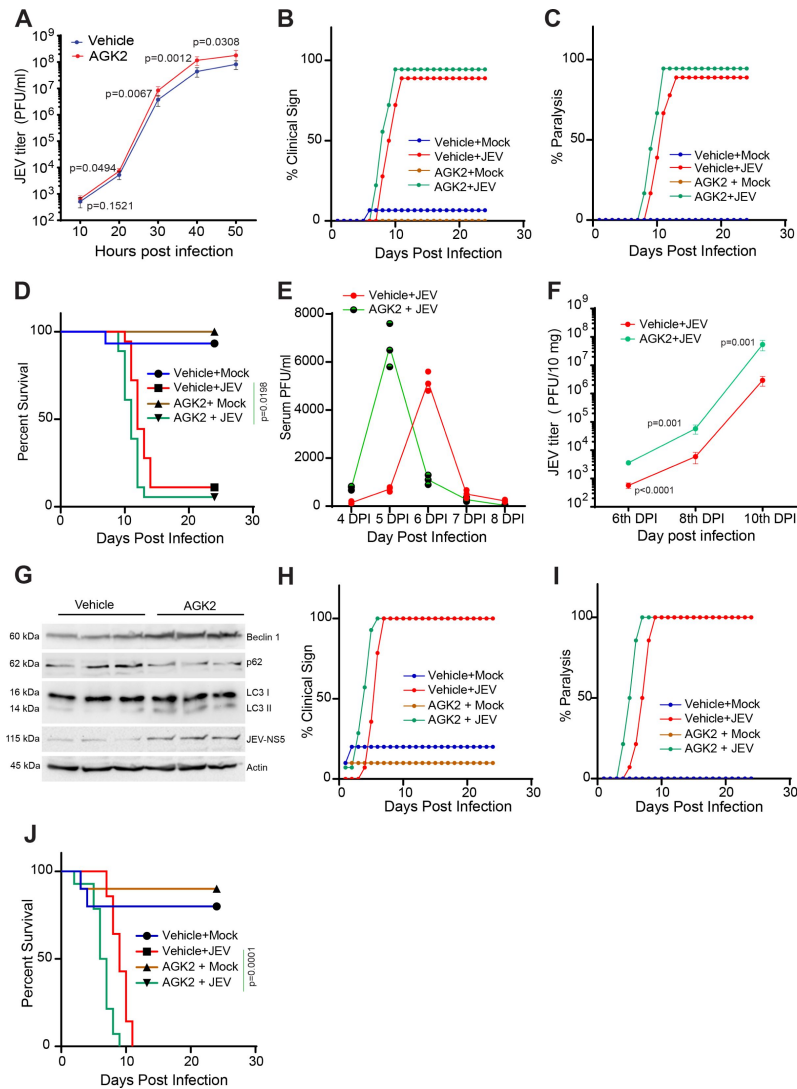


Figure 3.

AGK2 treatment exacerbated clinical manifestation in JEV-infected mice through induction of autophagy.

(A) Graph displaying the viral PFU levels at the various hpi of Vehicle or AGK2 treated neuro2a cells infected with JE-Virus. n=3 independent experiments; triplicates were used in each experiment. P values are calculated using Student's t-test. (B-F) 10-day-old WT mice were inoculated with JE virus of 10^5 PFUs by the foot-pad route; AGK2 treatment started from 1st dpi onwards for every day up to 24th dpi through the intraperitoneal (IP) route. (B) Graph representing the percent animals expressing clinical signs at various dpi in vehicle or AGK2 treated JE virus infected mice; n=15-18 mice per group. (C) Graph depicting the percent of animals displaying paralysis at various dpi in JE virus-infected mice upon vehicle or AGK2 treatment. n=15-18 mice per group. (D) The percentage of mice surviving after JE virus infection upon vehicle or AGK2 treatment at various dpi is depicted in the Kaplan-Meier survival curve. Statistical analysis was done using Log rank (Mantel-Cox) test; n=15-18 mice per group. Graph representing JE virus load in (E) serum (PFU/ml) (n=3 mice per dpi) and (F) brain (PFU/10mg) (n=5 mice per dpi), upon vehicle or AGK2 treatment in JE virus-infected mice at various dpi. The p-value was calculated using Student's t-test. (G) Representative immunoblot displaying differences in autophagy markers in the brains of the vehicle or AGK2 treated mice at 8th dpi of JE virus infection (n=3 mice each group). (H-J) 10-day-old Wild-Type mice were inoculated with 3×10^3 PFUs of JEV virus by an intra-cerebral route of inoculation, and AGK2 treatment started from first dpi for every day up to 24th dpi by the IP route of inoculation; n=10-14 mice each group. Graph displaying the percent of animals showing (H) clinical signs and (I) paralysis, at various dpi in JE virus-infected mice upon vehicle or AGK2 treatment. (n=10-14 mice each group). (J) The survival of JE virus-infected mice treated with vehicle or AGK2 up to 24th dpi in the Kaplan-Meier survival percentage curve. The p-value was calculated using the Log-rank (Mantel-Cox) test. (n=10-14 mice per group)

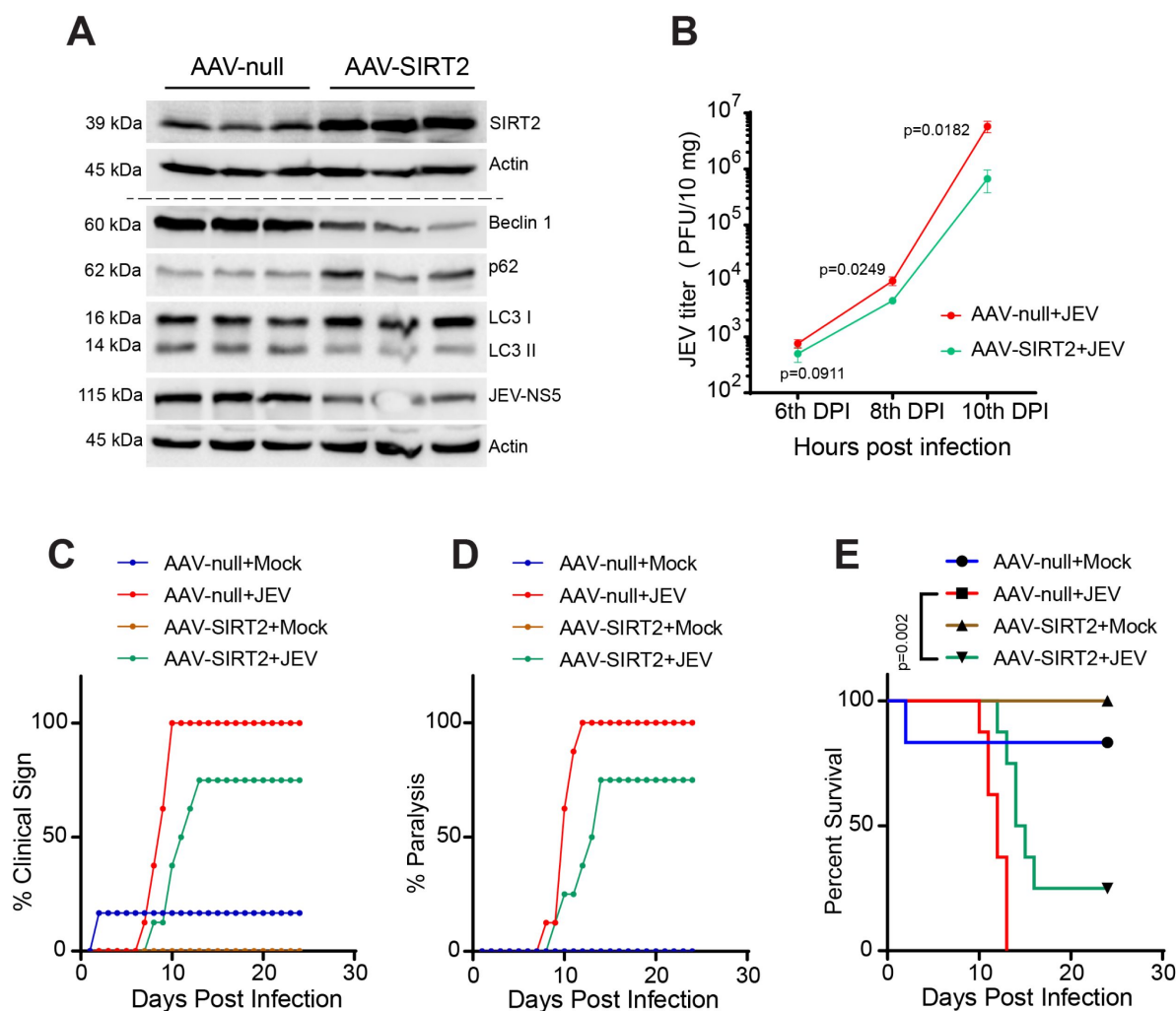


Figure 4.

SIRT2 gene therapy using the AAV vector protects against JEV infection in mice.

(A-E) 10-day-old wild-type mice were inoculated with 10^5 PFUs of JEV through the foot-pad route, and 10^{15} copies of AAV-null or AAV-SIRT2 treatment started from first dpi and 4th dpi through the IP route. (A) Representative immune-blot displaying changes in SIRT2 protein and autophagy markers in the brain of AAV-null or AAV-SIRT2 treated wild-type mice infected with JEV at 8th dpi (n=3 mice each group). (B) The graph represents JE virus PFUs in the brain of JEV-infected wild-type mice, with AAV-null or AAV-SIRT2 treatment; n=3 mice per dpi, respectively. The p-value was calculated using Student's t-test. The graph represents the percentage of animals exhibiting (C) clinical signs and (D) paralysis in mock or JE virus-infected wild-type mice treated with AAV-null or AAV-SIRT2 at various dpi (n=6-8 mice in each group). (E) Kaplan-Meier survival curve indicating the survival percentage of mock or JE virus-infected wild-type mice, with AAV-null or AAV-SIRT2 treatment at various dpi up to 24th dpi (n=6-8 mice in each group). The p-value was calculated using the Log-rank (Mantel-Cox) test.

SIRT2 deficiency regulates NF- κ B transcriptional factor activity to enhance the inflammatory response upon JEV infection in mice brains

To understand the pathways deregulated upon JEV infection, we performed global mRNA-sequencing of mock and virus-infected WT mice brains. Our global transcriptome analysis demonstrated that JEV infection led to the deregulation of several genes including those involved in immune response (**Figure 5A** [↗](#); Figure S2A). We also analyzed the transcription factors associated with the differentially expressed genes (**Figure 5B** [↗](#); Figure S2B). According to our KEGG pathway analysis, cytokine and its receptor interaction was the most significantly upregulated pathway (**Figure 5C** [↗](#)). The transcription factor corresponding to the most significantly upregulated genes was NF- κ B (**Figure 5D** [↗](#)). We also identified downregulated pathways and the associated transcription factors (**Figure 5E** [↗](#)-**5F** [↗](#)). Further, we analyzed transcriptomic data to determine the expression levels of approximately 400 genes regulated by NF- κ B transcription factor⁵⁰ [↗](#). Our analysis revealed that several NF- κ B target cytokines and chemokines were transcriptionally upregulated in JEV-infected mice brains (**Figure 5G** [↗](#)-**5H** [↗](#); Figure S3A, S3B). Further, a comparison of transcriptomic data between JEV-infected WT and SIRT2-KO mice brains revealed a significant increase in several NF- κ B target inflammatory cytokines and chemokine genes (**Figure 5I** [↗](#)-**5J** [↗](#); Figure S4A, S4B). Next, we validated the global transcriptomics results using quantitative PCR. We observed increased transcript levels of NF- κ B target inflammatory cytokines and chemokine genes in JEV-infected mice brains. These transcripts were further increased in the JEV-infected SIRT2-KO mice brains (**Figure 5K** [↗](#)-**5N** [↗](#); Figure S5A-S5C). During JEV infection, neuronal damage occurs (i) directly by virus replication in neurons and (ii) directly by JEV-induced increased cytokines in the brain⁵¹⁻⁵³ [↗](#). Therefore, the increased susceptibility of SIRT2-KO mice to JEV infection could be attributed to both cytokine release due to hyperactivation of NF- κ B and increased replication of the JEV in the brain.

SIRT2 mediated acetylation of NF- κ B regulates JEV replication through NF- κ B-Beclin1 axis of autophagy

Since our findings indicate that SIRT2 regulates JEV replication in neuro2a cells and mice brains (**Figure 2A** [↗](#)-**2B** [↗](#); Figure S1A-S1B; **Figure 2F** [↗](#); **Figure 2K** [↗](#); **Figure 3A** [↗](#); **Figure 3F** [↗](#)) with an increase in NF- κ B target gene expression in JEV-infected SIRT2-KO mice brain (**Figure 5A** [↗](#)-**5N** [↗](#); Figure S5A-S5C), we checked if SIRT2 directly interacts with the NF- κ B and regulate its activity upon JEV infection. To investigate SIRT2-NF- κ B protein-protein interaction, we immunoprecipitated NF- κ B from mock or JEV-infected neuro2a cells. We identified SIRT2 interacts with NF- κ B, and the interaction is reduced upon JEV infection (**Figure 6A** [↗](#)). Further, JEV infection increased acetylation levels of NF- κ B (**Figure 6A** [↗](#)). Next, we performed a reporter assay to evaluate the transcriptional activity of NF- κ B. In line with transcriptome analysis, we found that JEV infection increased NF- κ B transcriptional activity (**Figure 6B** [↗](#)). Interestingly, SIRT2 deficiency enhanced JEV-infection mediated NF- κ B transcriptional activity (**Figure 6C** [↗](#)) and increased the viral load in neuro2a cells (**Figure 2B** [↗](#)). Conversely, SIRT2 overexpression significantly reduced the JEV-infection-induced increase in NF- κ B transcriptional activity (**Figure 6D** [↗](#)). It also reduced the viral load in neuro2a cells (Figure S1A, S1B). Taken together, these findings reveal that JEV infection negatively regulates SIRT2 levels, resulting in a concomitant increase in acetylated NF- κ B mediated virus replication, and SIRT2 overexpression reverts this phenotype. Next, we attempted to identify the cellular signaling pathways related to NF- κ B target genes that may be involved in JEV replication. We compared the transcriptome profiles of JEV-infected WT and SIRT2-KO mice brains and carried out gene network analysis. We observed that NF- κ B mediated inflammation and autophagy were among the key axes that were deregulated in JEV-infected SIRT2-KO brains compared to JEV-infected WT mice brains (**Figure 6E** [↗](#)). Notably,

NF- κ B is established to induce autophagy by transcriptional upregulation of beclin-1⁵⁴. Our previous study has identified a crucial role for autophagy in JEV replication¹. Therefore, we explored the possibility of the SIRT2-NF- κ B-beclin-1 axis of autophagy in JEV infection. We observed that JEV infection significantly upregulated the transcript levels of beclin-1 (**Figure 6F**), in concordance with the increased NF- κ B transcriptional activity (**Figure 6B**). Additionally, SIRT2-KD also enhanced the transcript levels of beclin-1 (**Figure 6G**) and NF- κ B transcriptional activity (**Figure 6C**). Conversely, SIRT2-OE significantly reduced the transcript levels of beclin-1 (**Figure 6H**) and NF- κ B transcriptional activity (**Figure 6D**) and caused a reduction of viral load in neuro2a cells (Figure S1A-S1B). Further, we found that neuro2a cells infected with JEV showed increased autophagy as indicated by autophagy markers such as increased beclin-1, increased LC3II and reduced p62 (**Figure 6I**; Figure S6A). In line with our *in vitro* findings, JEV-infected mice brains also showed increased autophagy through increased beclin-1, LC3II, and reduced p62 concomitant with reduced SIRT2 levels (**Figure 6J**; Figure S6B). Further, we found that SIRT2 knockdown as well as chemical inhibition using AGK2 increased beclin-1 and autophagy in JEV-infected neuro2a cells (**Figure 6K**; Figure S6C; **Figure 6L**; Figure S6D) and mice brains (**Figure 3F**-**3G**). Moreover, JEV-infected SIRT2-KO mice showed increased beclin-1 and autophagy compared to their WT counterparts (**Figure 2E**; **Figure 6M**; Figure S6E). In contrast, SIRT2 overexpression significantly reduced autophagy and inhibited viral multiplication in neuro2a cells (Figure S1A, S1B; **Figure 6N**; Figure S6F). Moreover, we observed that AAV mediated SIRT2 gene therapy down-regulated the JEV-induced autophagy, which is critical for JEV replication, thereby exhibiting an antiviral role in the JEV-infected mice (**Figure 4A**). These results indicate that SIRT2 regulates JEV replication through the NF- κ B-beclin-1-autophagy axis.

SIRT2 deficiency promotes JEV replication in mice through NF- κ B activity

Previous studies indicate that NF- κ B activity is negatively regulated by NF- κ B inhibitor, I κ B, and this negative regulation is released by phosphorylation of I κ B^{42,43}. Thus, we treated the JEV-infected neuro2a cells with Bay-11, an inhibitor of I κ B α phosphorylation, and measured NF- κ B activity using a luciferase activity-based reporter assay. We observed a significant reduction in JEV-induced NF- κ B activity (**Figure 7A**) and a reduction in the viral load in neuro2a cells upon Bay-11 treatment (**Figure 7B**), indicating that I κ B α phosphorylation-mediated activation of NF- κ B activity is required for JEV replication. Moreover, Bay-11 further increased NF- κ B activity and JEV load in SIRT2 depleted neuro2a cells as compared to controls (**Figure 7C**, **7D**). In contrast, SIRT2 overexpression remarkably reduced JEV-induced NF- κ B transcription factor activity and JEV load which were reduced further upon Bay-11 treatment (**Figure 7E**, **7F**). Interestingly, Bay-11 significantly reduced SIRT2-KD mediated increase in NF- κ B transcriptional activity under JEV-infected conditions (**Figure 7C**). Similarly, Bay-11 significantly reduced viral load which was observed upon SIRT2-KD in neuro 2a cells (**Figure 7D**). Overall, these results reveal that SIRT2 exerts an anti-viral effect by deacetylating and inhibiting NF- κ B activity, which is required for JEV replication.

Subsequently, to determine whether SIRT2 regulates JEV replication in mice brains through NF- κ B, JEV-infected (through foot pad route of infection) WT and SIRT2-KO mice were treated with BAY-11. Here, we observed that BAY-11 treatment in JEV-infected WT mice reduced the viral load in both serum and brain, thereby reducing the clinical outcomes including paralysis, and enhancing survival compared to vehicle-treated JEV-infected WT mice (**Figure 8A**-**8E**). Furthermore, to study the therapeutic potential of BAY-11, we administered BAY-11 to the JEV-infected mice after the onset of JEV clinical signs. Even when administered after the onset of JEV clinical signs in mice, BAY-11 significantly reduced paralysis (Figure S7A), improved the survival percentage of mice infected with JEV (Figure S7B), and reduced the JEV load in the mice brains (Figure S7C). These results indicate that I κ B α phosphorylation mediated activation of NF- κ B is required for JEV

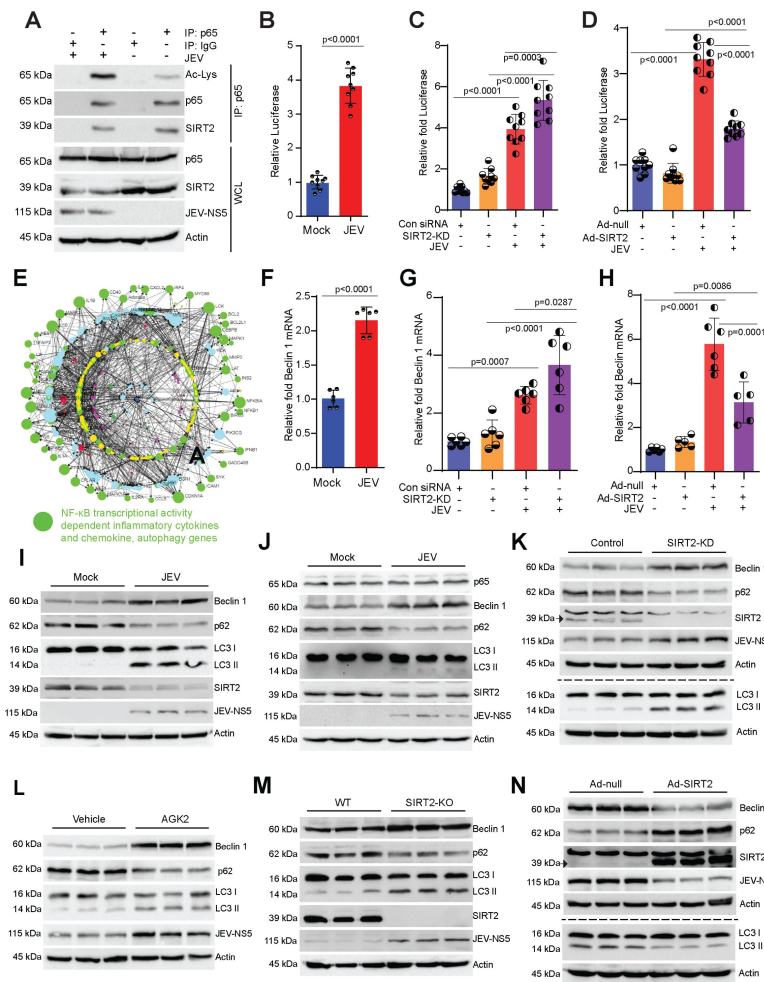


Figure 6.

SIRT2 and NF-κB interaction mediated acetylation status of NF-κB regulates the JEV replication.

(A) Immunoprecipitation of p65 at 30 hpi with mock or JE virus (1 MOI) infected neuro2a cells demonstrating p65 and SIRT2 interaction. (B) NF-κB activity was assayed at 30 hpi using luciferase reporter in mock or JE virus (1 MOI) infected neuro2a cells (n=3 independent experiments, triplicates were used in each experiment; The p-value was calculated using Student's t-test). Relative fold change in the luciferase reporter activity of NF-κB at 30 hpi of JE virus infection in (C) control or SIRT2-KD; (D) Ad-null or Ad-SIRT2; treated conditions; n=3 independent experiments, triplicates were used in each experiment; Statistical analysis using one-way ANOVA. (E) The network of signaling pathways represents differential expression genes that are regulated by NF-κB transcriptional activity in the WT or SIRT2^{-/-} mice brain upon JEV infection (n=3 mice each group). The qPCR analysis representing the relative fold changes in the beclin-1 expression levels in (F) JE virus-infected (1 MOI) neuro2a cells; (G) JEV (1 MOI) infected control or SIRT2-KD neuro2a cells; and (H) JEV (1 MOI) infected Ad-null or Ad-SIRT2 treated neuro2a cells at 30 hpi (n=3 independent experiments, triplicates were used in each experiment; The p-value was calculated using (B and F) Student's t-test and (C-D and G-H) one-way ANOVA. (I) Representative immuno-blotting images displaying differences in autophagy markers at 30 hpi in the JE virus-infected neuro2a cells. (n=3 independent experiments; duplicates were used in each experiment). (J) Representative immuno-blotting images displaying an increase in autophagy markers at 8 dpi in the JE virus-infected WT mice brain (n=6 mice in each group). (K) Representative immuno-blotting images displaying differences in autophagy markers at 30 hpi in the JE virus-infected, control or SIRT2-KD neuro2a cells (n=3 independent experiments; duplicates were used in each experiment). (L) Representative immuno-blotting images displaying change in autophagy markers at 30 hpi in the JE virus-infected, vehicle or AGK2-treated neuro2a cells (n=3 independent experiments; duplicates were used in each experiment). (M) Representative immuno-blotting images displaying differences in autophagy markers at 8 dpi in the WT or SIRT2^{-/-} mice brains upon JEV infection (n=6 mice in each group). (N) Representative immuno-blotting images displaying changes in autophagy markers at 30 hpi in the JE virus-infected neuro2a cells upon Ad-null or Ad-SIRT2 treatment (n=3 independent experiments; duplicates were used in each experiment). Statistical analysis of all immune-blotting experiments was performed using Student's t-test.

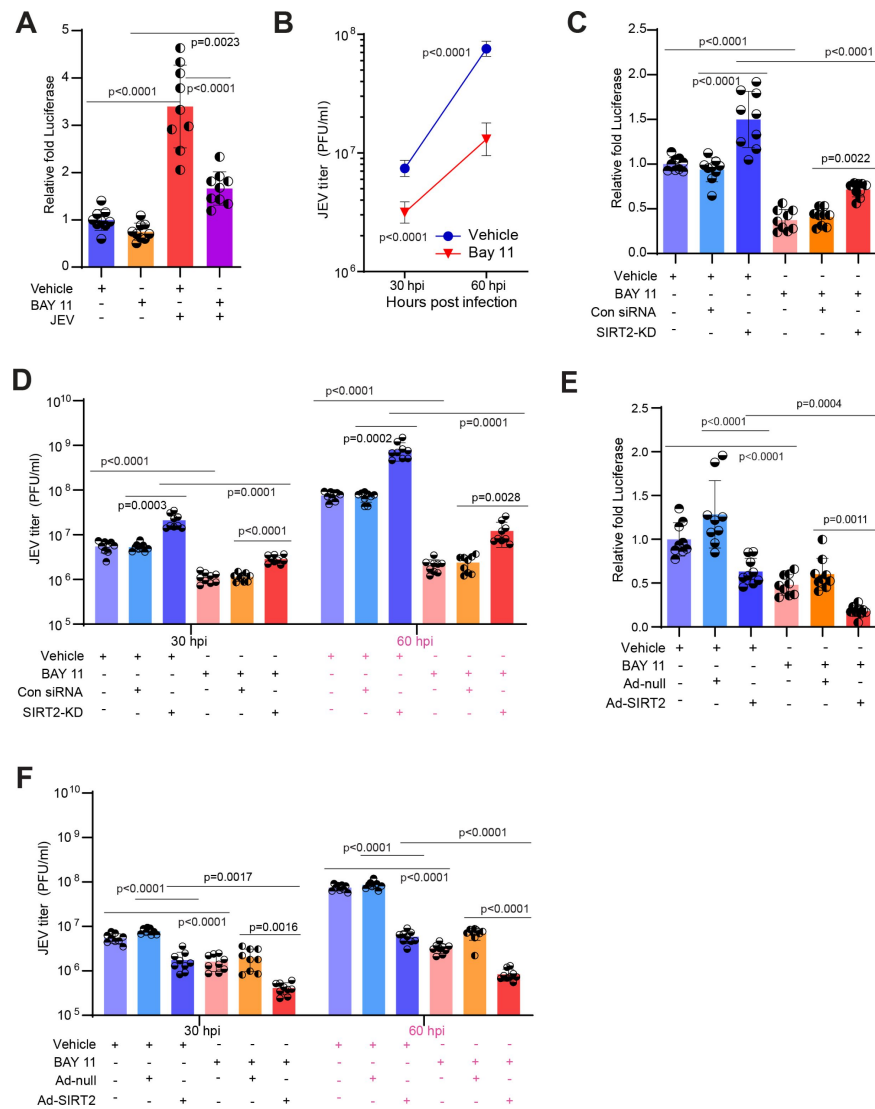


Figure 7.

SIRT2 exerts the antiviral effect by inhibiting NF-κB transcriptional activity.

(A) Graph displaying NF-κB activity assessed at 30 hpi using luciferase reporter assay in mock or JE virus (1 MOI) infected neuro2a cells treated with vehicle or Bay-11. Results are expressed as fold change relative to the vehicle-treated mock control group. (n=3 independent experiments, triplicates were used in each experiment; p-values are calculated using one-way ANOVA. (B) Graph corresponding to changes in viral PFUs at 30 and 60 hpi in JE virus (1 MOI) infected neuro2a cells with vehicle or Bay-11 treatment. n=3 independent experiments, triplicates were used in each experiment; The p-value was calculated using Student's t-test. (C) Graphical representations of NF-κB activity evaluated by luciferase reporter assays in JE virus (1 MOI) infected neuro2a cells treated with vehicle or Bay-11 in Control siRNA or SIRT2 specific siRNA transfected neuro2a cells; Results are depicted as fold change relative to vehicle-treated non-transfected control groups. n=3 independent experiments, triplicates were used in each experiment; Statistical analysis using one-way ANOVA. (D) Graph displaying changes in viral PFUs at 30 hpi in JEV infected neuro2a cells upon transfection with Control siRNA or SIRT2 specific siRNA and further treatment with vehicle or BAY-11. (n=3 independent experiments, triplicates were used in each experiment); p-values are calculated using one-way ANOVA. (E) Graphical representations of NF-κB activity evaluated by luciferase reporter assays in JE virus (1 MOI) infected neuro2a cells treated with vehicle or Bay-11 in Ad-null or Ad-SIRT2 treated neuro2a cells; Results are depicted as fold change relative to vehicle-treated non-transfected control groups. n=3 independent experiments, triplicates were used in each experiment; Statistical analysis using one-way ANOVA. (F) Graph displaying changes in viral PFUs at 30 hpi in JEV infected neuro2a cells upon treatment with Ad-null or Ad-SIRT2 and further treatment with vehicle or BAY-11. (n=3 independent experiments, triplicates were used in each experiment); p-values are calculated using one-way ANOVA.

replication in mice brains. Besides this, BAY-11 treatment in JEV-infected SIRT2-KO mice rescues the exacerbated clinical signs, paralysis, and decline in survival rates observed in the JEV-infected SIRT2-KO mice (**Figure 8A–8C**). Additionally, BAY-11 treatment reduced the increased viral load observed in JEV-infected SIRT2-KO mice serum and brain (**Figure 8D, 8E**). Collectively, our results suggest that SIRT2 deficiency in mice exacerbates JEV induced clinical signs, paralysis, and causes decline in survival rates through the activation of the NF- κ B transcription factor.

Discussion

While vaccines play an important role in preventing virus infection, antivirals are needed to complement the limitations of vaccines. Identifying host-directed antivirals is important as they reduce the resistance acquired by virus mutations^{55–58}. This present study describes SIRT2 as a host-directed antiviral factor against JEV, a public health threat due to the lack of specific complement antivirals in clinical practice despite the availability of vaccines. Notably, the existing studies have shown that SIRT2 knockdown increases the replication of a few DNA and RNA viruses in cell culture¹⁴. The present study elucidates that SIRT2 genetic deficiency or SIRT2 chemical inhibitors enhance JEV replication in cell culture and mouse models, while SIRT2 overexpression displays antiviral characteristics. Moreover, the present study established that SIRT2 acts as a host-directed antiviral by inhibiting JEV-induced autophagy by regulating the NF- κ B-Beclin-1-autophagy axis, thereby reducing virus replication.

SIRT2 is established to transcriptionally regulate several cellular signalling pathways through deacetylating transcriptional factors^{54,59–65}. In the case of DNA viruses that replicate in the nucleus, SIRT2 regulates transcriptional factors p53⁶⁶, and deacetylates G3BP, thereby regulating cGAS-STING-IRF3 transcriptional activity⁴⁸. Besides this, SIRT2 was reported to transcriptionally regulate autophagy by regulating transcriptional factors such as FoxO1 and TFEB^{67,68}. The present study reveals that NF- κ B deacetylation through SIRT2-NF- κ B interaction inhibits virus-induced autophagy. Mechanistically, SIRT2-mediated deacetylation of NF- κ B inhibits the transcriptional activity of the NF- κ B, thereby inhibiting the transcription of Beclin-1 and subsequent downstream autophagy, which is crucial for virus replication. The NF- κ B transcriptional activity regulates approximately 400 genes^{54,64,69}. On the other hand, Beclin-1 is reported to maintain the persistent activity of NF- κ B in human T cell leukaemia virus type 1 transformed T lymphocytes⁷⁰. Previous studies have reported that Beclin-1 regulates flavivirus-induced autophagy by viral non-structural protein 1 (NS1)⁷¹ and mosquito saliva-specific protein allergen-1⁷², indicating that Beclin-1 is needed for autophagy induction to favour flavivirus replication. Along these lines, our previous study has found that JEV infection enhances virus replication by inducing autophagy through the transcriptional factor PTEN¹. Our findings unveil that JEV infection downregulates SIRT2, which causes increased acetylation of NF- κ B, leading to the activation of the NF- κ B pathway.

Furthermore, we observed that the activated NF- κ B pathway by JEV infection induces autophagy through Beclin-1, thereby enhancing virus replication. Previous studies suggest both pro- and anti-inflammatory roles for SIRT2 in different pathologies. SIRT2 inhibition with AGK2 conferred protection against NF- κ B and MAPK activation mediated inflammatory response in the liver of mice model of thioacetamide-induced acute liver failure⁷³. Similarly, in other studies, including the lethal septic model and LPS-induced renal inflammation, SIRT2 inhibition attenuated inflammation^{74,75}. On the other hand, the anti-inflammatory role of SIRT2 is reported in traumatic brain injury and neuropathic pain^{76,77}. However, our findings show that SIRT2 depletion or chemical inhibition exacerbates inflammatory response and prognosis in mice models of JEV infection whereas SIRT2 gene therapy confers protection to mice against JEV infection. Previous findings suggest that the severity of JEV infection is largely manifested by inflammatory cytokine and chemokine hyperactivation^{53,78–80}. In the current study, we

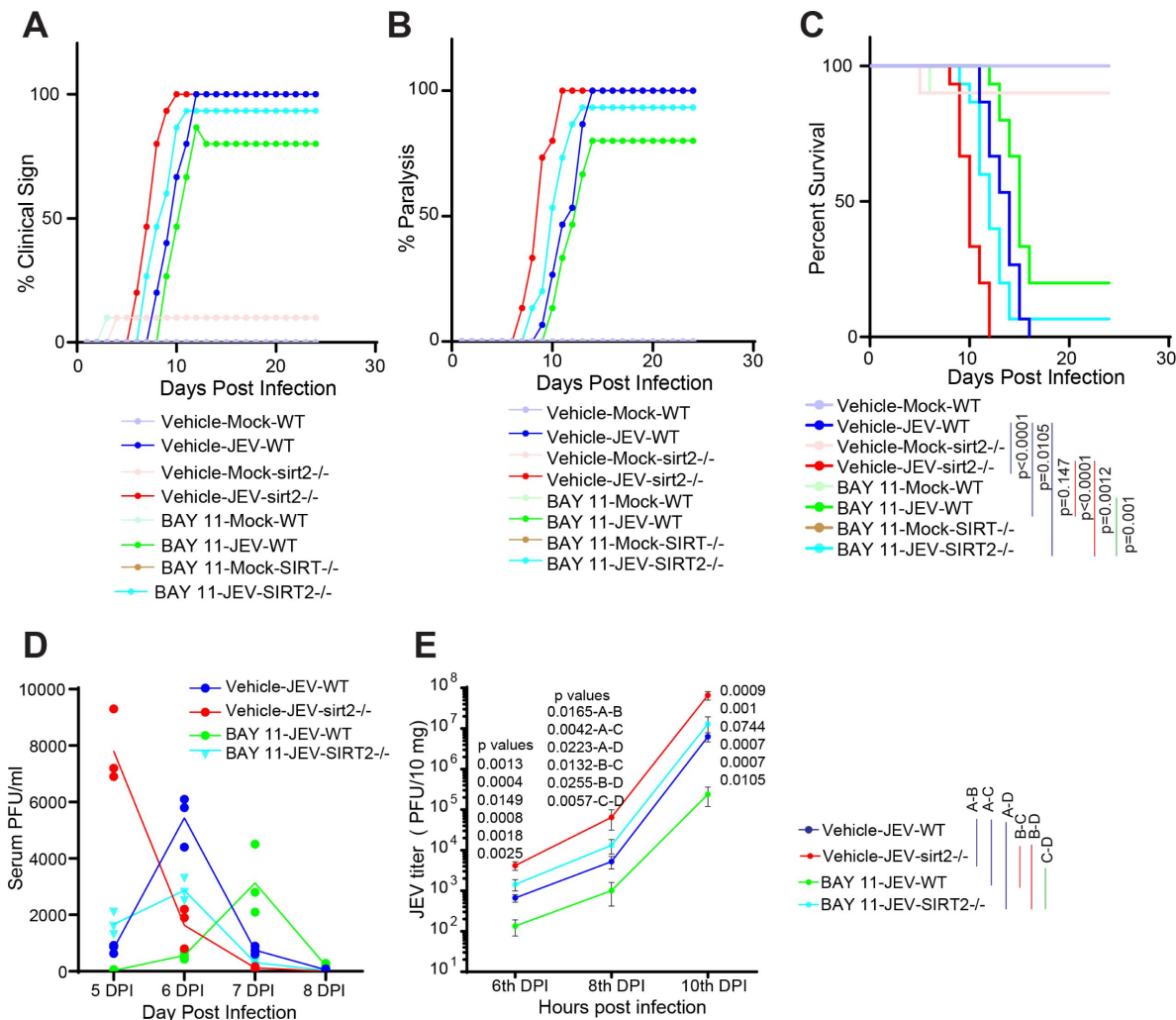


Figure 8.

SIRT2 regulates the NF-κB mediated antiviral activity in JE virus-infected mice.

(A-E) 10-day-old WT or SIRT2^{-/-} mice were inoculated with 10⁵ PFUs of JEV through the foot-pad route, and vehicle or BAY-11 treatment started from first dpi through the IP route for every day up to 24th dpi. The graph represents the percentage of WT or SIRT2^{-/-} animals exhibiting (A) clinical signs (B) paralysis in JE virus-infected vehicle or BAY-11 treated mice at various dpi (n=10-15 mice in each group). (C) Kaplan-Meier survival curve indicating the survival percentage of JE virus-infected, WT or SIRT2^{-/-} mice, with vehicle or BAY-11 treatment at various dpi up to 24th dpi (n=10-15 mice each group). The p-value was calculated using the Log-rank (Mantel-Cox) test. The graph represents JE virus PFUs in the (D) serum and (E) brain of infected WT or SIRT2^{-/-} mice, with vehicle or BAY-11 treatment; n=3 and 5 mice per dpi, respectively. The p-value was calculated using Student's t-test.

observed that the exacerbated clinical outcome in JEV-infected SIRT2-KO mice was mediated by NF- κ B transcriptional activity-dependent hyperactivation of inflammatory cytokines and chemokines.

Our findings establish that SIRT2 negatively regulates NF- κ B transcriptional activity and exhibits antiviral activity by restricting virus replication and reducing disease severity. We believe that SIRT2 gene therapy could be taken forward to the next level of clinical studies, which may improve the treatment of JEV infection. Our findings stress the potential therapeutic implications for RNA viral infections that utilize the evolutionarily conserved mechanism of the autophagy axis for favouring virus replication and inflammatory cytokines and chemokines hyperactivation for pathogenesis.

Materials and Methods

Experimental animals

In the present study, the SIRT2^{+/+} and SIRT2^{-/-} mice were procured from the Jackson Laboratory (B6.129-Sirt2tm1.1Fwa/J, RRID: IMSR_JAX:012772). The mice were housed and bred in the institute's central animal facility at Indian Institute of Science, Bengaluru, India. The mice were maintained in individually ventilated cages, with a 12-hour light/dark cycle. Food and water were provided ad libitum. All experiments were conducted after approvals from the institutional animal ethics committee and the guidelines of the Committee for the Purpose of Control and Supervision of Experiments on Animals, Government of India. Further, all the studies followed as per the U. S. National Institutes of Health (NIH Publication 2011) published 8th edition of Care and Use of Laboratory Animals.

JEV viral strain

Japanese encephalitis virus (JEV, Vellore strain-P20778) used in the study was a kind gift from Professor P. N. Rangarajan, Division of Biological Science, Department of Biochemistry, Indian Institute of Science (IISc), Bengaluru, India.

Method details

Mice model for JEV infection

The 10-day-old mice were infected with JEV by two routes, including foot-pad route and intra-cerebral infection at 10^5 and 3×10^3 PFU, respectively, and continuously observed and scored for the clinical signs, paralysis, and survival every day for 24 days of post-infection. Inhibitors AGK2 and BAY-11 were administered intraperitoneally at 2.5 and 2.5 mg/kg/day, respectively.

Cell culture, Cell transfection, and drug treatment

The *in vitro* studies were performed on neuro2a cells. The cells were cultured in high glucose DMEM supplemented with 10% foetal bovine serum (FBS) and an antimycotic-antibiotic mix and grown and maintained in a humidified incubator at 37°C and 5% CO₂. The siRNA-based SIRT2 knockdown and adenovirus-based overexpression of SIRT2 was carried out at 70-80% confluency as described previously^{63,81,82}. Briefly, siRNA transfection was carried out using Lipofectamine RNAiMAX per the manufacturer's protocol. In the case of adenovirus transduction, SIRT2 or null adenovirus, propagated and quantified in the HEK 293T cell line, was used with equal MOIs to infect the neuro2a cells. AGK2 and BAY 11-7082 were used to inhibit SIRT2 and IKK inhibitors at 10 μ M and 2.5 μ M concentrations, respectively.

JEV propagation and viral plaque assay

The BHK-21 cell line was used to propagate JEV and measure the plaque-forming units (PFU) as described previously¹. At a density of 0.2×10^5 BHK-21 cells per well, cells were seeded in six-well plates, and infected with ten-fold serial dilutions of JEV virus for 1 hour for the virus entry. After an hour, the uninfected viruses were removed by washing wells thrice using phosphate-buffered saline (PBS). Then cells were covered using 3% low-melting agarose in culture DMEM with 10% FBS and antibiotic-antimycotic. Further, it was allowed at room temperature for 15 minutes to solidify. After solidification, incubated in a humidified incubator at 37°C and 5% of CO₂ for 36-48 hours. Then the cells were examined microscopically after fixing using 10% formaldehyde for 10-20 minutes at room temperature and stained with a solution of 1.5% crystal violet. Then, the excess strains were removed by washing thrice with MilliQ water, and the plaques were counted using white light. According to the plague counts and virus serial dilution used, PFU was calculated per mL as follows: $PFU = N \times DF/V$ (DF is virus dilution factor, N is number of plaques, and V is the volume of the inoculum).

Dual-Luciferase reporter assay

To study the NF-κB activity, the cells (neuro2a) were co-transfected with plasmids containing NF-κB-Luciferase reporter and Renilla-Luciferase reporter with pRL-CMV using lipofectamine 3000. While NF-κB-Luciferase reporter activity corresponded to NF-κB activity, Renilla-Luciferase activity was used to normalise transfection. The cell lysis and assay were done in accordance with the Dual-Luciferase Reporter kit Promega, and Pharmingen Moonlight 3010, BD Biosciences, was used to measure the luminescence.

Cyto-immunofluorescence microscopy

Cyto-immunofluorescence microscopy was carried out as described previously¹. In brief, the cells were cultured on the coverslips and subjected to mentioned treatments. At the study time point, the cells were fixed by incubating with 10% formaldehyde for about 15-20 minutes at room temperature (RT). The fixed cells were washed thrice with PBS and then permeabilization was carried out using 0.25% Triton X-100 diluted in PBS by incubating at RT for 10-15 min. The permeabilized cells were blocked for 20 to 30 minutes with 1% BSA dissolved in PBS with 0.1% tween and 22.52 mg/mL glycine. The blocked cells were then incubated with diluted primary antibody at 4°C overnight. Post-incubation, the cells were washed thrice and incubated with diluted secondary antibodies conjugated with Alexa Fluor for an hour at RT. The nuclear stain Hoechst 33342 was also incubated with a secondary antibody. After secondary incubation, the coverslips were washed and mounted on a glass slide with a Fluoromount-G Aqueous mount. The cells were imaged using a confocal microscope Zeiss LSM 880.

Immunoprecipitation assay

Immunoprecipitation (IP) subsequently immunoblotting was done as described previously¹. In brief, the neuro2a cells, after aspiration of media, were washed thrice with PBS and cell lysates were prepared in RIPA cell lysis buffer. The protein levels were quantified using Bradford, and about 500 µg protein was used for IP. The lysate containing the specified protein levels was incubated with 2 µg of specific or control (IgG) antibodies rotating at 5 rpm overnight. After overnight incubation, washed protein A/G magnetic beads were added to the lysate containing antibodies and incubated at 4°C for 2 hours. Post-incubation, the entire mixture was passed through the column, washed thrice using PBS, and finally, elution was carried out using Laemmli buffer. The IP was measured by immunoblotting.

Real-time PCR

For the real-time PCR, TRI-reagent was used to extract RNA from the cells/tissues according to the manufacturer's guidelines. Then, the extracted RNAs were quantified in nanodrop, and 500 ng was used for cDNA synthesis utilizing the iScript cDNA Synthesis Kit. The real-time PCR was carried out using Quant studio 6 flex using the qPCR master mix with SYBR-Green.

Immuno-blotting

Immunoblotting was performed as described previously¹. In Brief, the cell/tissue lysates were prepared using RIPA buffer with phosphatase inhibitor, deacetylase inhibitors and protease inhibitor cocktail. The tissue samples were homogenized with liquid nitrogen and then collected in RIPA buffer. The lysates were centrifuged for 15 minutes at 12,000 rpm. The supernatants were collected separately and used for western blotting. The protein quantities were measured by using a Bradford reagent. An equal amount of protein for each sample was loaded on 12% SDS-PAGE and resolved followed by transfer to a PVDF membrane (0.45µm) by overnight transfer method. Upon completion of the transfer, the PVDF membrane (0.45µm) was incubated in 5% skimmed milk in Tris-buffered saline with Tween-20 (TBST) for an hour at RT. Then, the PVDF membrane was washed using TBST for thrice, three minutes each and incubated with the primary antibody of the mentioned dilutions at 4°C for overnight. Post-incubation, the unbound antibodies were washed off with three TBST washes for three minutes each. The membrane was incubated with appropriate secondary antibodies conjugated to horseradish peroxidase for an hour at RT. The chemiluminescence was detected with ECL reagents, and the image was attained in a chemiluminescence imager.

Generation of adeno-associated virus

Mouse full-length SIRT2 cDNA was cloned into the AAV-CMV-GFP (Plasmid #67634)⁸³ by replacing GFP to generate AAV-CMV-mSIRT2. The pAAV2/9n (Plasmid #112865)⁸⁴ was used RepCap plasmids by inserting WVLPSSG between amino acids 588 and 589, corresponding to AAV9 VP1⁸⁵ in this plasmid by using the QuikChange Site-Directed Mutagenesis Kit⁸⁶. The pAdDeltaF6 (Plasmid #112867)⁸⁷ was used as an AAV helper plasmid. The AAV-CMV-mSIRT2, pAdDeltaF6, and WVLPSSG-inserted pAAV2/9n were co-transfected in the HEK 293T cell line to generate the AAV vector to deliver SIRT2 into mice's brains. The control null-AAV vector was generated by deleting the SIRT2 in the AAV-CMV-mSIRT2 and co-transfected with pAdDeltaF6, and WVLPSSG-inserted pAAV2/9n in the HEK 293T cell line. The resulting AAV vector was purified by ultracentrifugation as per the previous protocol^{83,88,89}. The AAV genomic DNA copy number was determined by qPCR. Equal copies of purified SIRT2-AAV and null-AAV were used to infect the mice through the intraperitoneal route.

Global mRNA sequencing and analysis

Briefly, the total RNA was isolated for transcriptome sequencing using TRI-reagent from the mock or JEV-infected SIRT2^{+/+} and SIRT2^{-/-} mice brains. The quality control was done with the 4150 Tape Station-System, Agilent-G2992AA with RNA ScreenTape-System, Agilent, 5067-5576. Samples with the above 8 RNA integrity number (RINe) values were utilized to further subsequent downstream processing. To enrich for mRNA in the samples, NEBNext Poly (A) mRNA magnetic isolation module was used and library preparation was carried out using 500ng of total RNA as per the manufacturer's protocol, New England Biolabs-E7490 was employed. After the library preparation, the global mRNA sequencing was carried out in an Illumina-HiSeq sequencer. The obtained sequences were validated in FastQC (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). Then, fastp v0.20⁹⁰ was used to remove the adapter sequences and low-quality reads. Next, STAR v2⁹¹ with Mus musculus database was used to map the genes. The feature-counts⁹² was used for mapped read counts and gene expression values. The genes that showed consistent differential

expression when normalised using edgeR⁹³. The gene network analysis was performed using SiGNaling Network Open Resource with SIGNOR 2.0. The KEGG pathways enrichment and TRRUST Transcription Factors enrichment analysis were carried out using Enrichr^{94–96}.

Statistical analysis

The statistical analysis and graphs included in the study were carried out in the GraphPad Prism 8.0, GraphPad Software Inc., USA. The mean±standard deviation was used to highlights in the graphs. The p-value was calculated using the student's t-test, and one-way ANOVA, a <0.05 p-value was considered as statistically significant. For survival curve was determined using the Kaplan-Meier curve. The p values in the same were calculated using the Log-rank (Mantel-Cox) test. The ZEN image analysis software was used to analyse the confocal images. The western blots were quantified using ImageJ software.

Data and code availability

The global mRNA-seq data from the mock or JEV-infected SIRT2^{+/+} and SIRT2^{-/-} mice brain has been submitted in the NCBI (PRJNA1001148 and PRJNA1091868). These data are accessible publicly from the date of manuscript publication.

Key resources used in the study

Designation	Reagent / Resource type	Source	Catalog number/ Identifier
SIRT2-KO mice	Mice	Jackson Laboratories, USA,	B6.129-Sirt2tm1.1Fwa/J, RRID: IMSR_JAX:012772
Japanese Encephalitis Virus	Virus	Professor P. N. Rangarajan, Department of Biochemistry, IISc, Bangalore, India	Vellore strain (P20778),
Anti-SIRT2	Antibody	Sigma-Aldrich	S8447
Anti- NF-κB p65	Antibody	Cell Signaling Technology	(D14E12); 8242
Anti-Beclin-1	Antibody	Cell Signaling Technology	3738
Anti-Acetylated-Lysine	Antibody	Cell Signaling Technology	9814
Anti-LC3B	Antibody	Cell Signaling Technology	3868
Anti-p62	Antibody	Cell Signaling Technology	5114
Anti-β-Actin (HRP Conjugated)	Antibody	Sigma-Aldrich	A3854
Anti-JEV-NS5	Antibody	Invitrogen	PA5-111977
Anti- Rabbit HRP	Antibody	Cell Signaling Technology	7074
Anti-mouse HRP	Antibody	Cell Signaling Technology	7076
cDNA synthesis kit	Commercial assay/kit	BioRad	
High capacity cDNA reverse transcription kit	Commercial assay/kit	Thermo Fisher Scientific	4368814
iTaq universal SYBR® GREEN super mix	Chemical/Reagent	BioRad	L001751 A
Protein Assay Dye Reagent Concentrate	Chemical/Reagent	BioRad	L007096 D
Lipofectamine ®2000 Reagent	Commercial assay/kit	Thermo Fisher Scientific	11668-019

Lipofectamine ®3000 Reagent	Commercial assay/kit	Thermo Fisher Scientific	L3000-015
KAPA PROBE FAST	Commercial assay/kit	Sigma-Aldrich	kk4703
QuantStudio™ 6 Flex Real-Time PCR System	Equipment	Thermo fisher Scientific	
LSM 880 confocal microscope	Equipment	Zeiss	
ChemiDoc™ Touch Imaging System	Equipment	BioRad	
Luminometer	Equipment	Turner Machine, TD-20/20	
Hoechst 33342	Chemical/Reagent	Thermo fisher Scientific	H3570
Fetal Bovine Serum	Cell culture reagent	Thermo fisher Scientific	10500064
Dulbecco's Modified Eagle's Medium- High glucose	Cell culture media	Sigma-Aldrich	AT007-50L
ProLong Gold Antifade Mountant with DAPI	Chemical /Reagent	Thermo fisher Scientific	P36931
cOmplete™, Mini Protease Inhibitor Cocktail	Chemical /Reagent	Sigma-Aldrich, ROCHE	11836153001
Dual-Luciferase® Reporter Assay System	Commercial assay or kit	Promega	E1910
Amersham Hybond PVDF membrane	Miscellaneous	GE Healthcare	10600023
Clarity ECL Western Blotting Substrate	Chemical /Reagent	BioRad	5060
Trizol	Chemical /Reagent	Takara	9108
Lipofectamine™ RNAiMAX Transfection Reagent	Chemical /Reagent	Thermo fisher Scientific	13778150
Antibiotic-Antimycotic mixture	Chemical /Reagent	Thermo fisher Scientific	15240062
BAY 11-7082	Chemical /Reagent	Sigma-Aldrich	196870
AGK2 (CAS 304896-28-4)	Chemical /Reagent	Sigma-Aldrich	566324
Biophotometer	Equipment	Eppendorf	
Centrifuge	Equipment	Eppendorf	

q-PCR primers

Name	Forward	Reverse
Beclin 1	5'CTCTGAAACTGGACACGAGC3'	5'CTGGGCTGTGGTAAGTAATGG3'
GAPDH	5'TGCAGTGGCAAAGTGGAGATT3'	5'TTGAATTTGCCGTGAGTGGAA3'
IL23a	5'TGTGCCCCGTATCCAGTGTG3'	5'GAAGATGTCAGAGTCAAGCAG3'
CXCL10	5'TCATCCACCGCTGAGAGACC3'	5'GAAAATGACGGCAGCACTTGG3'
Cd69	5'GGCTGTGTTAATAGTGGTCCTC3'	5'TTGCAGGTAGCAACATGGTGG3'
CCL5	5'TTTCCTACCTCTCCCTCG3'	5'CGACTGCAAGATTGGAGCACT3'
CXCL11	5'TGTAATTTACCCGAGTAACGGC3'	5'CACCTTTGTCGTTTATGAGCCTT3'
CD80	5'ATACACCACTCCTCAAGTTTCC3'	5'TAACGGCAAGGCAGCAATACC3'

JEV qPCR primer and Probe

JEV	Forward	JEV-RT-F-5'-TGGTCCAYAGGGAATGGTT-3',
	Reverse	JEV-RT-R-AAGAGCAACAACGGACTGTTT-3',
	Probe	JEV-RT-P-5'FAM-TTTGAAGAGGCGCACGCCAC-Tamra-3'

Acknowledgements

We are grateful to Prof. P.N. Rangarajan, Department of Biochemistry, IISc, Bangalore, India, for gifting the JE virus. The PAD. received INSPIRE faculty (DST/INSPIRE/04/2016/001067) support for fellowship and research funds from the Department of Science and Technology, Government of India. P.A.D. received a research grant from the Science and Engineering Research Board (SERB) (CRG/2018/002192), Government of India.

Additional information

Author contributions

PAD and NRS conceived the idea of the study and planned all experimental conditions. PAD carried out almost all experiments. LD and VR assisted PAD in performing experiments. PAD, AKT and RSR were involved in the mice experiments. PAD and KM carried out the data arrangement. The first draft of the manuscript was written by PAD and KM. AS and SR engaged in data verification and manuscript review. The final version of the manuscript was written by PAD and NRS.

References

- 1 Desingu P. A., et al. (2023) **PARP1 inhibition protects mice against Japanese encephalitis virus infection** *Cell Rep* **42**:113103 <https://doi.org/10.1016/j.celrep.2023.113103> | Google Scholar
- 2 Desingu P. A., et al. (2016) **Pathogenic and Genotypic Characterization of a Japanese Encephalitis Virus Isolate Associated with Reproductive Failure in an Indian Pig Herd** *PLoS One* **11**:e0147611 <https://doi.org/10.1371/journal.pone.0147611> | Google Scholar
- 3 Desingu P. A., et al. (2017) **First Complete Genome Sequence of Genotype III Japanese Encephalitis Virus Isolated from a Stillborn Piglet in India** *Genome Announc* **5** <https://doi.org/10.1128/genomeA.01503-16> | Google Scholar
- 4 Gao X. M., et al. (2006) **Inhibition of mTOR reduces chronic pressure-overload cardiac hypertrophy and fibrosis** *J Hypertens* **24**:1663–1670 <https://doi.org/10.1097/01.hjh.0000239304.01496.83> | Google Scholar
- 5 (no date) <https://www.who.int/news-room/fact-sheets/detail/japanese-encephalitis>. no date <https://www.who.int/news-room/fact-sheets/detail/japanese-encephalitis>
- 6 Lauring A. S., Hodcroft E. B (2021) **Genetic Variants of SARS-CoV-2-What Do They Mean?** *JAMA* **325**:529–531 <https://doi.org/10.1001/jama.2020.27124> | Google Scholar
- 7 Dolan P. T., Whitfield Z. J., Andino R (2018) **Mapping the Evolutionary Potential of RNA Viruses** *Cell Host Microbe* **23**:435–446 <https://doi.org/10.1016/j.chom.2018.03.012> | Google Scholar
- 8 Duffy S (2018) **Why are RNA virus mutation rates so damn high?** *PLoS Biol* **16**:e3000003 <https://doi.org/10.1371/journal.pbio.3000003> | Google Scholar
- 9 Peck K. M., Lauring A. S (2018) **Complexities of Viral Mutation Rates** *J Virol* **92** <https://doi.org/10.1128/JVI.01031-17> | Google Scholar
- 10 Rack J. G., VanLinden M. R., Lutter T., Aasland R., Ziegler M (2014) **Constitutive nuclear localization of an alternatively spliced sirtuin-2 isoform** *J Mol Biol* **426**:1677–1691 <https://doi.org/10.1016/j.jmb.2013.10.027> | Google Scholar
- 11 Liu G., et al. (2017) **Loss of NAD-Dependent Protein Deacetylase Sirtuin-2 Alters Mitochondrial Protein Acetylation and Dysregulates Mitophagy** *Antioxid Redox Signal* **26**:849–863 <https://doi.org/10.1089/ars.2016.6662> | Google Scholar
- 12 Alqarni M. H., Foudah A. I., Muharram M. M., Labrou N. E (2021) **The Pleiotropic Function of Human Sirtuins as Modulators of Metabolic Pathways and Viral Infections** *Cells* **10** <https://doi.org/10.3390/cells10020460> | Google Scholar
- 13 Wu Q. J., et al. (2022) **The sirtuin family in health and disease** *Signal Transduct Target Ther* **7**:402 <https://doi.org/10.1038/s41392-022-01257-8> | Google Scholar

- 14 Koyuncu E., et al. (2014) **Sirtuins are evolutionarily conserved viral restriction factors** *mBio* **5** <https://doi.org/10.1128/mBio.02249-14> | Google Scholar
- 15 Wang Y., Yang J., Hong T., Chen X., Cui L (2019) **SIRT2: Controversy and multiple roles in disease and physiology** *Ageing Res Rev* **55**:100961 <https://doi.org/10.1016/j.arr.2019.100961> | Google Scholar
- 16 Baeken M. W (2023) **Sirtuins and their influence on autophagy** *J Cell Biochem* <https://doi.org/10.1002/jcb.30377> | Google Scholar
- 17 Wang L., Ou J. J (2018) **Regulation of Autophagy by Hepatitis C Virus for Its Replication** *DNA Cell Biol* **37**:287–290 <https://doi.org/10.1089/dna.2017.4115> | Google Scholar
- 18 Chan S. T., Ou J. J (2017) **Hepatitis C Virus-Induced Autophagy and Host Innate Immune Response** *Viruses* **9** <https://doi.org/10.3390/v9080224> | Google Scholar
- 19 Sun M. X., et al. (2012) **Porcine reproductive and respiratory syndrome virus induces autophagy to promote virus replication** *Autophagy* **8**:1434–1447 <https://doi.org/10.4161/auto.21159> | Google Scholar
- 20 Chen Q., et al. (2012) **Induction of autophagy enhances porcine reproductive and respiratory syndrome virus replication** *Virus Res* **163**:650–655 <https://doi.org/10.1016/j.virusres.2011.11.008> | Google Scholar
- 21 Prentice E., Jerome W. G., Yoshimori T., Mizushima N., Denison M. R (2004) **Coronavirus replication complex formation utilizes components of cellular autophagy** *J Biol Chem* **279**:10136–10141 <https://doi.org/10.1074/jbc.M306124200> | Google Scholar
- 22 Feizi N., et al. (2017) **Autophagy induction regulates influenza virus replication in a time-dependent manner** *J Med Microbiol* **66**:536–541 <https://doi.org/10.1099/jmm.0.000455> | Google Scholar
- 23 Wang R., et al. (2019) **Autophagy Promotes Replication of Influenza A Virus In Vitro** *J Virol* **93** <https://doi.org/10.1128/JVI.01984-18> | Google Scholar
- 24 Zhang J., et al. (2018) **Flaviviruses Exploit the Lipid Droplet Protein AUP1 to Trigger Lipophagy and Drive Virus Production** *Cell Host Microbe* **23**:819–831 <https://doi.org/10.1016/j.chom.2018.05.005> | Google Scholar
- 25 Heaton N. S., Randall G (2010) **Dengue virus-induced autophagy regulates lipid metabolism** *Cell Host Microbe* **8**:422–432 <https://doi.org/10.1016/j.chom.2010.10.006> | Google Scholar
- 26 Cortese M., et al. (2017) **Ultrastructural Characterization of Zika Virus Replication Factories** *Cell Rep* **18**:2113–2123 <https://doi.org/10.1016/j.celrep.2017.02.014> | Google Scholar
- 27 Offerdahl D. K., Dorward D. W., Hansen B. T., Bloom M. E (2017) **Cytoarchitecture of Zika virus infection in human neuroblastoma and Aedes albopictus cell lines** *Virology* **501**:54–62 <https://doi.org/10.1016/j.virol.2016.11.002> | Google Scholar
- 28 Krejbich-Trotot P., et al. (2011) **Chikungunya triggers an autophagic process which promotes viral replication** *Virology* **432** <https://doi.org/10.1186/1743-422X-8-432> | Google Scholar

- 29 Klein K. A., Jackson W. T (2011) **Human rhinovirus 2 induces the autophagic pathway and replicates more efficiently in autophagic cells** *J Virol* **85**:9651–9654 <https://doi.org/10.1128/JVI.00316-11> | Google Scholar
- 30 Lee Y. R., Wang P. S., Wang J. R., Liu H. S (2014) **Enterovirus 71-induced autophagy increases viral replication and pathogenesis in a suckling mouse model** *J Biomed Sci* **21**:80 <https://doi.org/10.1186/s12929-014-0080-4> | Google Scholar
- 31 Bird S. W., Maynard N. D., Covert M. W., Kirkegaard K (2014) **Nonlytic viral spread enhanced by autophagy components** *Proc Natl Acad Sci U S A* **111**:13081–13086 <https://doi.org/10.1073/pnas.1401437111> | Google Scholar
- 32 Jackson W. T., et al. (2005) **Subversion of cellular autophagosomal machinery by RNA viruses** *PLoS Biol* **3**:e156 <https://doi.org/10.1371/journal.pbio.0030156> | Google Scholar
- 33 Alirezaei M., Flynn C. T., Wood M. R., Harkins S., Whitton J. L (2015) **Coxsackievirus can exploit LC3 in both autophagy-dependent and -independent manners in vivo** *Autophagy* **11**:1389–1407 <https://doi.org/10.1080/15548627.2015.1063769> | Google Scholar
- 34 Mohamud Y., et al. (2018) **Enteroviral Infection Inhibits Autophagic Flux via Disruption of the SNARE Complex to Enhance Viral Replication** *Cell Rep* **22**:3292–3303 <https://doi.org/10.1016/j.celrep.2018.02.090> | Google Scholar
- 35 Hou L., et al. (2014) **Nonstructural proteins 2C and 3D are involved in autophagy as induced by the encephalomyocarditis virus** *Virol J* **11**:156 <https://doi.org/10.1186/1743-422X-11-156> | Google Scholar
- 36 O'Donnell V., et al. (2011) **Foot-and-mouth disease virus utilizes an autophagic pathway during viral replication** *Virology* **410**:142–150 <https://doi.org/10.1016/j.virol.2010.10.042> | Google Scholar
- 37 Sun Y., et al. (2014) **Autophagy benefits the replication of Newcastle disease virus in chicken cells and tissues** *J Virol* **88**:525–537 <https://doi.org/10.1128/JVI.01849-13> | Google Scholar
- 38 Zhao J., He S., Minassian A., Li J., Feng P (2015) **Recent advances on viral manipulation of NF-kappaB signaling pathway** *Curr Opin Virol* **15**:103–111 <https://doi.org/10.1016/j.coviro.2015.08.013> | Google Scholar
- 39 Rahman M. M., McFadden G (2011) **Modulation of NF-kappaB signalling by microbial pathogens** *Nat Rev Microbiol* **9**:291–306 <https://doi.org/10.1038/nrmicro2539> | Google Scholar
- 40 Struzik J., Szulc-Dabrowska L (2018) **NF-kappaB Signaling in Targeting Tumor Cells by Oncolytic Viruses-Therapeutic Perspectives** *Cancers* **10** <https://doi.org/10.3390/cancers10110426> | Google Scholar
- 41 Santoro M. G., Rossi A., Amici C (2003) **NF-kappaB and virus infection: who controls whom** *EMBO J* **22**:2552–2560 <https://doi.org/10.1093/emboj/cdg267> | Google Scholar
- 42 Sun S. C (2012) **The noncanonical NF-kB pathway** *Immunol Rev* **246**:125–140 <https://doi.org/10.1111/j.1600-065X.2011.01088.x> | Google Scholar
- 43 Sun S. C (2017) **The non-canonical NF-kB pathway in immunity and inflammation** *Nat Rev Immunol* **17**:545–558 <https://doi.org/10.1038/nri.2017.52> | Google Scholar

- 44 Wu Q.-J., et al. (2022) **The sirtuin family in health and disease** *Signal Transduction and Targeted Therapy* **7**:402 <https://doi.org/10.1038/s41392-022-01257-8> | Google Scholar
- 45 Ramli N. S., et al. (2022) **Seroepidemiological Studies on Japanese Encephalitis: A Systematic Review** *Oman Med J* **37**:e366 <https://doi.org/10.5001/omj.2021.86> | Google Scholar
- 46 Budayeva H. G., Rowland E. A., Cristea I. M (2016) **Intricate Roles of Mammalian Sirtuins in Defense against Viral Pathogens** *J Virol* **90**:5–8 <https://doi.org/10.1128/JVI.03220-14> | Google Scholar
- 47 Zhang S., Wang J., Wang L., Aliyari S., Cheng G (2022) **SARS-CoV-2 virus NSP14 Impairs NRF2/HMOX1 activation by targeting Sirtuin 1** *Cell Mol Immunol* **19**:872–882 <https://doi.org/10.1038/s41423-022-00887-w> | Google Scholar
- 48 Li Y., et al. (2023) **SIRT2 negatively regulates the cGAS-STING pathway by deacetylating G3BP1** *EMBO Rep* **24**:e57500 <https://doi.org/10.15252/embr.202357500> | Google Scholar
- 49 Yu H. B., et al. (2018) **AGK2, A SIRT2 Inhibitor, Inhibits Hepatitis B Virus Replication In Vitro And In Vivo** *Int J Med Sci* **15**:1356–1364 <https://doi.org/10.7150/ijms.26125> | Google Scholar
- 50 Wang Y., et al. (2021) **KLK11 promotes the activation of mTOR and protein synthesis to facilitate cardiac hypertrophy** *BMC Cardiovasc Disord* **21**:266 <https://doi.org/10.1186/s12872-021-02053-y> | Google Scholar
- 51 Cao S., et al. (2011) **Japanese encephalitis Virus wild strain infection suppresses dendritic cells maturation and function, and causes the expansion of regulatory T cells** *Virology* **8**:39 <https://doi.org/10.1186/1743-422X-8-39> | Google Scholar
- 52 Lannes N., Summerfield A., Filgueira L (2017) **Regulation of inflammation in Japanese encephalitis** *J Neuroinflammation* **14**:158 <https://doi.org/10.1186/s12974-017-0931-5> | Google Scholar
- 53 Shukla M., Garg A., Dhole T. N., Chaturvedi R (2023) **Exaggerated levels of some specific TLRs, cytokines and chemokines in Japanese encephalitis infected BV2 and neuro 2A cell lines associated with worst outcome** *Virology* **20**:16 <https://doi.org/10.1186/s12985-023-01966-8> | Google Scholar
- 54 Copetti T., Bertoli C., Dalla E., Demarchi F., Schneider C (2009) **p65/RelA modulates BECN1 transcription and autophagy** *Mol Cell Biol* **29**:2594–2608 <https://doi.org/10.1128/MCB.01396-08> | Google Scholar
- 55 Kaufmann S. H. E., Dorhoi A., Hotchkiss R. S., Bartenschlager R (2018) **Host-directed therapies for bacterial and viral infections** *Nat Rev Drug Discov* **17**:35–56 <https://doi.org/10.1038/nrd.2017.162> | Google Scholar
- 56 Dwek R. A., Bell J. I., Feldmann M., Zitzmann N (2022) **Host-targeting oral antiviral drugs to prevent pandemics** *Lancet* **399**:1381–1382 [https://doi.org/10.1016/S0140-6736\(22\)00454-8](https://doi.org/10.1016/S0140-6736(22)00454-8) | Google Scholar
- 57 Olivet J., et al. (2022) **A systematic approach to identify host targets and rapidly deliver broad-spectrum antivirals** *Mol Ther* **30**:1797–1800 <https://doi.org/10.1016/j.ymthe.2022.02.015> | Google Scholar

- 58 Cakir M., Obernier K., Forget A., Krogan N. J (2021) **Target Discovery for Host-Directed Antiviral Therapies: Application of Proteomics Approaches** *mSystems* **6**:e0038821 <https://doi.org/10.1128/mSystems.00388-21> | Google Scholar
- 59 Yang X., et al. (2017) **Sirtuin 2 regulates cellular iron homeostasis via deacetylation of transcription factor NRF2** *J Clin Invest* **127**:1505–1516 <https://doi.org/10.1172/JCI88574> | Google Scholar
- 60 Bae N. S., Swanson M. J., Vassilev A., Howard B. H (2004) **Human histone deacetylase SIRT2 interacts with the homeobox transcription factor HOXA10** *J Biochem* **135**:695–700 <https://doi.org/10.1093/jb/mvh084> | Google Scholar
- 61 Zhang X., Azhar G., Wei J. Y (2017) **SIRT2 gene has a classic SRE element, is a downstream target of serum response factor and is likely activated during serum stimulation** *PLoS One* **12**:e0190011 <https://doi.org/10.1371/journal.pone.0190011> | Google Scholar
- 62 Jing E., Gesta S., Kahn C. R (2007) **SIRT2 regulates adipocyte differentiation through FoxO1 acetylation/deacetylation** *Cell Metab* **6**:105–114 <https://doi.org/10.1016/j.cmet.2007.07.003> | Google Scholar
- 63 Sarikhani M., et al. (2018) **SIRT2 deacetylase represses NFAT transcription factor to maintain cardiac homeostasis** *J Biol Chem* **293**:5281–5294 <https://doi.org/10.1074/jbc.RA117.000915> | Google Scholar
- 64 Lin F., et al. (2015) **XIAP and cIAP1 amplifications induce Beclin 1-dependent autophagy through NFkappaB activation** *Hum Mol Genet* **24**:2899–2913 <https://doi.org/10.1093/hmg/ddv052> | Google Scholar
- 65 Rothgiesser K. M., Erener S., Waibel S., Luscher B., Hottiger M. O (2010) **SIRT2 regulates NF-kappaB dependent gene expression through deacetylation of p65 Lys310** *J Cell Sci* **123**:4251–4258 <https://doi.org/10.1242/jcs.073783> | Google Scholar
- 66 Wu D. Q., et al. (2022) **SIRT2 Promotes HBV Transcription and Replication by Targeting Transcription Factor p53 to Increase the Activities of HBV Enhancers and Promoters** *Front Microbiol* **13**:836446 <https://doi.org/10.3389/fmicb.2022.836446> | Google Scholar
- 67 Zhao Y., et al. (2010) **Cytosolic FoxO1 is essential for the induction of autophagy and tumour suppressor activity** *Nat Cell Biol* **12**:665–675 <https://doi.org/10.1038/ncb2069> | Google Scholar
- 68 Wang L., et al. (2020) **Down Regulation of SIRT2 Reduced ASS Induced NSCLC Apoptosis Through the Release of Autophagy Components via Exosomes** *Front Cell Dev Biol* **8**:601953 <https://doi.org/10.3389/fcell.2020.601953> | Google Scholar
- 69 Prakash P., Gorfe A. A (2022) **Determinants of Membrane Orientation Dynamics in Lipid-Modified Small GTPases** *JACS Au* **2**:128–135 <https://doi.org/10.1021/jacsau.1c00426> | Google Scholar
- 70 Chen L., Liu D., Zhang Y., Zhang H., Cheng H (2015) **The autophagy molecule Beclin 1 maintains persistent activity of NF-kappaB and Stat3 in HTLV-1-transformed T lymphocytes** *Biochem Biophys Res Commun* **465**:739–745 <https://doi.org/10.1016/j.bbrc.2015.08.070> | Google Scholar

- 71 Lu Z. Y., et al. (2020) **Dengue Nonstructural Protein 1 Maintains Autophagy through Retarding Caspase-Mediated Cleavage of Beclin-1** *Int J Mol Sci* **21** <https://doi.org/10.3390/ijms21249702> | Google Scholar
- 72 Sun P., et al. (2020) **A mosquito salivary protein promotes flavivirus transmission by activation of autophagy** *Nat Commun* **11**:260 <https://doi.org/10.1038/s41467-019-14115-z> | Google Scholar
- 73 Jiao F. Z., et al. (2019) **Protective role of AGK2 on thioacetamide-induced acute liver failure in mice** *Life Sci* **230**:68–75 <https://doi.org/10.1016/j.lfs.2019.05.061> | Google Scholar
- 74 Jung Y. J., et al. (2015) **SIRT2 Regulates LPS-Induced Renal Tubular CXCL2 and CCL2 Expression** *J Am Soc Nephrol* **26**:1549–1560 <https://doi.org/10.1681/asn.2014030226> | Google Scholar
- 75 Zhao T., et al. (2015) **Selective Inhibition of SIRT2 Improves Outcomes in a Lethal Septic Model** *Curr Mol Med* **15**:634–641 <https://doi.org/10.2174/156652401507150903185852> | Google Scholar
- 76 Yuan F., et al. (2016) **SIRT2 inhibition exacerbates neuroinflammation and blood-brain barrier disruption in experimental traumatic brain injury by enhancing NF- κ B p65 acetylation and activation** *J Neurochem* **136**:581–593 <https://doi.org/10.1111/jnc.13423> | Google Scholar
- 77 Zhang Y., Chi D (2018) **Overexpression of SIRT2 Alleviates Neuropathic Pain and Neuroinflammation Through Deacetylation of Transcription Factor Nuclear Factor-Kappa B** *Inflammation* **41**:569–578 <https://doi.org/10.1007/s10753-017-0713-3> | Google Scholar
- 78 Chen C. J., et al. (2010) **Glial activation involvement in neuronal death by Japanese encephalitis virus infection** *J Gen Virol* **91**:1028–1037 <https://doi.org/10.1099/vir.0.013565-0> | Google Scholar
- 79 Saxena S. K., et al. (2023) **Antiviral and anti-inflammatory activity of novel belladonna formulation against Japanese encephalitis virus via inhibition of p65 nuclear translocation and TNF-alpha mediated NF-kB signaling** *Biotechnol Genet Eng Rev* :1–23 <https://doi.org/10.1080/02648725.2023.2166258> | Google Scholar
- 80 Bian P., et al. (2017) **Mesenchymal stem cells alleviate Japanese encephalitis virus-induced neuroinflammation and mortality** *Stem Cell Res Ther* **8**:38 <https://doi.org/10.1186/s13287-017-0486-5> | Google Scholar
- 81 Sarikhani M., et al. (2018) **SIRT2 deacetylase regulates the activity of GSK3 isoforms independent of inhibitory phosphorylation** *eLife* **7** <https://doi.org/10.7554/eLife.32952> | Google Scholar
- 82 Sarikhani M., et al. (2018) **SIRT2 regulates oxidative stress-induced cell death through deacetylation of c-Jun NH(2)-terminal kinase** *Cell Death Differ* **25**:1638–1656 <https://doi.org/10.1038/s41418-018-0069-8> | Google Scholar
- 83 Xiong W., MacColl Garfinkel A. E., Li Y., Benowitz L. I., Cepko C. L (2015) **NRF2 promotes neuronal survival in neurodegeneration and acute nerve damage** *J Clin Invest* **125**:1433–1445 <https://doi.org/10.1172/JCI79735> | Google Scholar

- 84 (no date) **James M. Wilson (Addgene plasmid # 112865 ; <http://n2t.net/addgene:112865> ; [RRID:Addgene_112865](http://n2t.net/addgene:112865)). no date** <http://n2t.net/addgene:112865>
- 85 Stanton A. C., et al. (2023) **Systemic administration of novel engineered AAV capsids facilitates enhanced transgene expression in the macaque CNS** *Med* **4**:31–50 <https://doi.org/10.1016/j.medj.2022.11.002> | Google Scholar
- 86 Ravi V., et al. (2019) **SIRT6 transcriptionally regulates global protein synthesis through transcription factor Sp1 independent of its deacetylase activity** *Nucleic Acids Res* **47**:9115–9131 <https://doi.org/10.1093/nar/gkz648> | Google Scholar
- 87 (no date) **James M. Wilson (Addgene plasmid # 112867 ; <http://n2t.net/addgene:112867> ; [RRID:Addgene_112867](http://n2t.net/addgene:112867)). no date** <http://n2t.net/addgene:112867>
- 88 Zhuo M., Mao J., David C. J (2023) **Protocol to package and concentrate adeno-associated virus serotype 8 for use in autochthonous mouse models of pancreatic cancer** *STAR Protoc* **4**:102108 <https://doi.org/10.1016/j.xpro.2023.102108> | Google Scholar
- 89 Ohba K., Mizukami H (2023) **Protocol for producing an adeno-associated virus vector by controlling capsid expression timing** *STAR Protoc* **4**:102542 <https://doi.org/10.1016/j.xpro.2023.102542> | Google Scholar
- 90 Chen S., Zhou Y., Chen Y., Gu J. (2018) **fastp: an ultra-fast all-in-one FASTQ preprocessor** *Bioinformatics* **34**:i884–i890 <https://doi.org/10.1093/bioinformatics/bty560> | Google Scholar
- 91 Dobin A., et al. (2013) **STAR: ultrafast universal RNA-seq aligner** *Bioinformatics* **29**:15–21 <https://doi.org/10.1093/bioinformatics/bts635> | Google Scholar
- 92 Liao Y., Smyth G. K., Shi W (2014) **featureCounts: an efficient general purpose program for assigning sequence reads to genomic features** *Bioinformatics* **30**:923–930 <https://doi.org/10.1093/bioinformatics/btt656> | Google Scholar
- 93 Robinson M. D., McCarthy D. J., Smyth G. K (2010) **edgeR: a Bioconductor package for differential expression analysis of digital gene expression data** *Bioinformatics* **26**:139–140 <https://doi.org/10.1093/bioinformatics/btp616> | Google Scholar
- 94 Chen E. Y., et al. (2013) **Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool** *BMC Bioinformatics* **14**:128 <https://doi.org/10.1186/1471-2105-14-128> | Google Scholar
- 95 Kuleshov M. V., et al. (2016) **Enrichr: a comprehensive gene set enrichment analysis web server 2016 update** *Nucleic Acids Res* **44**:W90–97 <https://doi.org/10.1093/nar/gkw377> | Google Scholar
- 96 Xie Z., et al. (2021) **Gene Set Knowledge Discovery with Enrichr** *Curr Protoc* **1**:e90 <https://doi.org/10.1002/cpz1.90> | Google Scholar

Author information

Perumal Arumugam Desingu

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India,
Department of Virus Epidemiology, Vector Dynamics & Public Health, Institute of Advanced
Virology, Trivandrum, India
ORCID iD: [0000-0003-0011-3948](https://orcid.org/0000-0003-0011-3948)

For correspondence: padesingu@iav.res.in

Lavanya Dindi

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India

Krishnega Murugasamy

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India

Ankit Kumar Tamta

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India

Venketsubbu Ramasubbu

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India

Sukanya Raghu

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India

Amarjeet Shrama

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India

Raju S Rajmani

Centre for Infectious Disease Research, Indian Institute of Science, Bengaluru, India

Nagalingam R Sundaresan

Department of Microbiology and Cell Biology, Indian Institute of Science, Bengaluru, India
ORCID iD: [0000-0003-1770-5616](https://orcid.org/0000-0003-1770-5616)

For correspondence: rsundaresan@iisc.ac.in

Editors

Reviewing Editor

Melody Man Hing Li

University of California, Los Angeles, Los Angeles, United States of America

Senior Editor

John Schoggins

The University of Texas Southwestern Medical Center, Dallas, United States of America

Reviewer #1 (Public review):

Summary:

Desingu et al. show that JEV infection reduces SIRT2 expression. Upon JEV infection, 10-day-old SIRT2 KO mice showed increased viral titer, more severe clinical outcomes, and reduced

survival. Conversely, SIRT2 overexpression reduced viral titer, clinical outcomes, and improved survival. Transcriptional profiling shows dysregulation of NF- κ B and expression of inflammatory cytokines. Pharmacological NF- κ B inhibition reduced viral titer. The authors conclude that SIRT2 is a regulator of JEV infection.

Strengths:

This paper is novel because sirtuins have been primarily studied for aging, metabolism, stem cells/regeneration. Their role in infection has not been explored until recently. Indeed, Barthez et al. showed that SIRT2 protects aged mice from SARS-CoV-2 infection (Barthez, Cell Reports 2025). Therefore, this is a timely and novel research topic. Mechanistically, the authors showed that SIRT2 suppresses the NF- κ B pathway. Interestingly, SIRT2 has also been shown recently to suppress other major inflammatory pathways, such as cGAS-STING (Barthez, Cell Reports 2025) and the NLRP3 inflammasome (He, Cell Metabolism 2020; Luo, Cell Reports 2019). Together, these findings support the emerging concept that SIRT2 is a master regulator of inflammation.

Weaknesses:

(1) Figures 2 and 3. Although SIRT2 KO mice showed increased viral titer, more severe clinical outcomes, and reduced survival upon JEV infection, the difference is modest because even WT mice exhibited very severe disease at this viral dose. The authors should perform the experiment using a sub-lethal viral dose for WT mice, to allow the assessment of increased clinical outcomes and reduced survival in KO mice.

(2) Figure 5K-N, the authors examined the expression of inflammatory cytokines in WT and SIRT2 KO cells upon JEV infection, in line with the dysregulation of NF- κ B. It has been shown recently that SIRT2 also regulates the cGAS-STING pathway (Barthez, Cell Reports 2025) and the NLRP3 inflammasome (He, Cell Metabolism 2020; Luo, Cell Reports 2019). Do you also observe increased IFN β , IL1 β , and IL18 in SIRT2 KO cells upon JEV infection? This may indicate that SIRT2 regulates systemic inflammatory responses and represents a potent protection upon viral infection. This is particularly important because in Figure 7F, the authors showed that SIRT2 overexpression reduced viral load even when NF- κ B is inhibited, suggesting that NF- κ B is not the only mediator of SIRT2 to suppress viral infection.

<https://doi.org/10.7554/eLife.106778.1.sa2>

Reviewer #2 (Public review):

The manuscript by Desingu et al., explores the role of SIRT2 in regulating Japanese Encephalitis Virus (JEV) replication and disease progression in rodent models. Using both an in vitro and an in vivo approach, the authors demonstrate that JEV infection leads to decreased SIRT2 expression, which they hypothesize is exploited by JEV for viral replication. To test this hypothesis, the authors utilize SIRT2 inhibition (via AGK2 or genetic knockout) and demonstrate that it leads to increased viral load and worsens clinical outcomes in JEV-infected mice. Conversely, SIRT2 overexpression via an AAV delivery system reduces viral replication and improves survival among infected mice. The study proposes a mechanism in which SIRT2 suppresses JEV-induced autophagy and inflammation by deacetylating NF- κ B, thereby reducing Beclin-1 expression (an NF- κ B-dependent gene) and autophagy, which the authors consider a pathway that JEV exploits for replication. Transcriptomic analysis further supports that SIRT2 deficiency leads to NF- κ B-driven cytokine hyperactivation. Additionally, pharmacological inhibition of NF- κ B using Bay 11 (an IKK inhibitor) results in reduced viral load and improved clinical pathology in WT and SIRT2 KO mice. Overall, the findings from Desingu et al. are generally supported by the data and suggest that targeting SIRT2 may serve as a promising therapeutic approach for JEV infection and potentially other RNA viruses that

SIRT2 helps control. However, the paper does fall short in some areas. Please see below for our comments to help improve the paper.

<https://doi.org/10.7554/eLife.106778.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

Desingu et al. show that JEV infection reduces SIRT2 expression. Upon JEV infection, 10-day-old SIRT2 KO mice showed increased viral titer, more severe clinical outcomes, and reduced survival. Conversely, SIRT2 overexpression reduced viral titer, clinical outcomes, and improved survival. Transcriptional profiling shows dysregulation of NF-KB and expression of inflammatory cytokines. Pharmacological NF-KB inhibition reduced viral titer. The authors conclude that SIRT2 is a regulator of JEV infection.

This paper is novel because sirtuins have been primarily studied for aging, metabolism, stem cells/regeneration. Their role in infection has not been explored until recently. Indeed, Barthez et al. showed that SIRT2 protects aged mice from SARS-CoV-2 infection (Barthez, Cell Reports 2025). Therefore, this is a timely and novel research topic. Mechanistically, the authors showed that SIRT2 suppresses the NF-KB pathway. Interestingly, SIRT2 has also been shown recently to suppress other major inflammatory pathways, such as cGAS-STING (Barthez, Cell Reports 2025) and the NLRP3 inflammasome (He, Cell Metabolism 2020; Luo, Cell Reports 2019). Together, these findings support the emerging concept that SIRT2 is a master regulator of inflammation.

Weaknesses:

(1) Figures 2 and 3. Although SIRT2 KO mice showed increased viral titer, more severe clinical outcomes, and reduced survival upon JEV infection, the difference is modest because even WT mice exhibited very severe disease at this viral dose. The authors should perform the experiment using a sub-lethal viral dose for WT mice, to allow the assessment of increased clinical outcomes and reduced survival in KO mice.

(2) Figure 5K-N, the authors examined the expression of inflammatory cytokines in WT and SIRT2 KO cells upon JEV infection, in line with the dysregulation of NF-kB. It has been shown recently that SIRT2 also regulates the cGAS-STING pathway (Barthez, Cell Reports 2025) and the NLRP3 inflammasome (He, Cell Metabolism 2020; Luo, Cell Reports 2019). Do you also observe increased IFN β , IL1 β , and IL18 in SIRT2 KO cells upon JEV infection? This may indicate that SIRT2 regulates systemic inflammatory responses and represents a potent protection upon viral infection. This is particularly important because in Figure 7F, the authors showed that SIRT2 overexpression reduced viral load even when NF-KB is inhibited, suggesting that NF-KB is not the only mediator of SIRT2 to suppress viral infection.

We thank the reviewer for the valuable recommendation. We are willing to conduct an experiment using a sub-lethal viral dose in wild-type (WT) mice to assess increased clinical outcomes and reduced survival in knockout (KO) mice, as recommended.

Furthermore, we acknowledge reviewers' comments that SIRT2 regulates systemic inflammatory responses and provides potent protection against viral infection. Additionally,

NF- κ B is not the only mediator of SIRT2's suppression of viral infection; other possible molecular mechanisms are also involved in this process.

Reviewer #2 (Public review):

The manuscript by Desingu et al., explores the role of SIRT2 in regulating Japanese Encephalitis Virus (JEV) replication and disease progression in rodent models. Using both an in vitro and an in vivo approach, the authors demonstrate that JEV infection leads to decreased SIRT2 expression, which they hypothesize is exploited by JEV for viral replication. To test this hypothesis, the authors utilize SIRT2 inhibition (via AGK2 or genetic knockout) and demonstrate that it leads to increased viral load and worsens clinical outcomes in JEV-infected mice. Conversely, SIRT2 overexpression via an AAV delivery system reduces viral replication and improves survival among infected mice. The study proposes a mechanism in which SIRT2 suppresses JEV-induced autophagy and inflammation by deacetylating NF- κ B, thereby reducing Beclin-1 expression (an NF- κ B-dependent gene) and autophagy, which the authors consider a pathway that JEV exploits for replication. Transcriptomic analysis further supports that SIRT2 deficiency leads to NF- κ B-driven cytokine hyperactivation. Additionally, pharmacological inhibition of NF- κ B using Bay 11 (an IKK inhibitor) results in reduced viral load and improved clinical pathology in WT and SIRT2 KO mice. Overall, the findings from Desingu et al. are generally supported by the data and suggest that targeting SIRT2 may serve as a promising therapeutic approach for JEV infection and potentially other RNA viruses that SIRT2 helps control. However, the paper does fall short in some areas. Please see below for our comments to help improve the paper.

We thank the reviewer for the valuable recommendation. We are willing to measure NF- κ B acetylation in AdSIRT2 JEV-infected cells compared to WT-infected cells, to verify that the acetylation of NF- κ B is truly linked to SIRT2 expression levels as per the reviewers' suggestion.

We are willing to conduct an experiment using a sub-lethal viral dose in wild-type (WT) mice to assess increased clinical outcomes and reduced survival in knockout (KO) mice, as recommended.

We are accepting the reviewer's suggestion that AGK2 can also inhibit other Sirtuins. Thus, to test the contribution of other Sirtuins, the experiment could be repeated using wild-type and Sirt2 KO mice. We are willing to conduct the AGK2 experiment using JEV-infected wild-type and Sirt2 knockout mice.

<https://doi.org/10.7554/eLife.106778.1.sa0>